

Design and Analysis of Second-Order Passive Filters



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ABSTRACT

This report investigates the behavior and design of second-order passive filters. Four configurations were built and analyzed: low pass, high pass and band pass. The first three circuits (low pass, high and band pass) were built in a single loop configuration, while the last filter (band pass) was built in a two-loop configuration. Transfer functions were derived analytically and verified using Multisim simulations. Selected circuits were constructed on a breadboard and measured experimentally. Theoretical cutoff frequencies and -3 dB points closely matched simulated and measured results, confirming expected filter behavior and stability characteristics.

INTRODUCTION

Filters are circuits designed to selectively allow signals within a desired frequency range while attenuating unwanted frequency components. They are commonly used in signal processing, communication systems, and instrumentation. Filters can be implemented as passive filters (as seen in this lab, as well as in lab 1) and active filters (as seen in lab 2), as well as composite (combinations of low, high, and bandpass filters). Filter are also classified by order, and second order specifically is addressed in this report. In this lab, the behavior of second order passive filter circuits (single and double looped) is analyzed by examining their bode plots, transfer functions and frequency responses in order to determine the filtering characteristics of the circuits.

REVIEW

Filters can be classified as active or passive. Electronic components can be classified as passive or active devices. Passive devices include resistors, capacitors, inductors, and diodes. These components do not provide a gain in power—they just store, dissipate, or direct energy. Active devices include transistors and op amps (operational amplifiers). These components require an external power source and are capable of providing amplification. When passive elements (such as an RC filter circuit) are combined with active elements (such as an op amp), this results in an active circuit.

Filters can also be classified by their order. A first-order circuit contains resistors and capacitors, *or* inductors, and has one zero and one pole. A second-order circuit contains two first-order circuits—resistance, plus two inductive or capacitive elements. Second order has two poles two zeros. Similarly, a third-order circuit consists of a second-order circuit plus a first-order circuit; resistance plus three inductive or capacitive elements, and three poles. Increasing the order of a filter produces a steeper attenuation beyond the cutoff frequency, which gets the circuit closer to the ideal goal of reducing unwanted frequency components outside of the desired bandwidth. The greater the order the steeper the attenuation past the cutoff frequency. To analyze filter behavior, the transfer function of the circuit is determined and analyzed for zero (numerator) and poles (denominator).

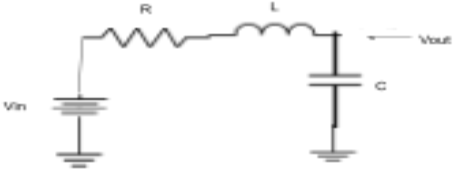
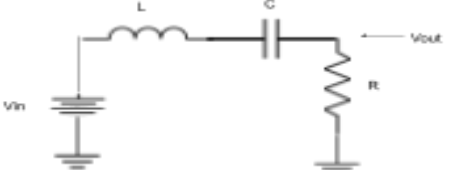
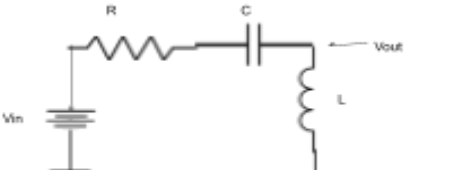
Filter type	Configuration of passive devices	Transfer function	Zeros	Roots
Low pass		$H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$	Zero at infinity	$s = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{2L^2} - \frac{1}{LC}}$
Band pass		$H(S) = \frac{\frac{R}{L}S}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$	Zero at zero	$s = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{2L^2} - \frac{1}{LC}}$
High Pass		$H(S) = \frac{s^2}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$	Double zero at zero	$s = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{2L^2} - \frac{1}{LC}}$

Table 1. Transfer function and analysis for single-loop second-order passive filters.

We set $H(s) = \frac{n(Z)}{n(P)}$ and then $n(Z) = 0$ as well as $n(p) = 0$. $n(Z)$ tells us filter behavior and $n(p)$ gives us a quadratic equation from $s^2 + (\frac{R}{L})s + (\frac{1}{LC}) = 0$ because it is in the form $ax^2 + bx + c$, so $a = 1$, $b = \frac{R}{L}$, $c = \frac{1}{LC}$, which gives the roots as $s = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{2L^2} - \frac{1}{LC}}$. Comparing to the standard second order equation of $\frac{1}{s^2 + 2\xi\omega_o s + \omega_o^2}$ gives us a term equivalent to ω_o^2 which can be used to solve for the value of the resistance. For critically dampened, the term $\frac{R^2}{2L^2} - \frac{1}{LC}$ from $\sqrt{\frac{R^2}{2L^2} - \frac{1}{LC}}$ must be equivalent to 0 in order for the poles to be identical and real. Therefore, $\frac{R^2}{2L^2} - \frac{1}{LC} = 0$ and $R = 2\sqrt{\frac{L}{C}}$ in addition to $R = 2L\omega_n \xi$ (from comparison to the second order standard equation).

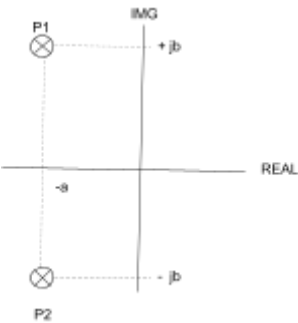
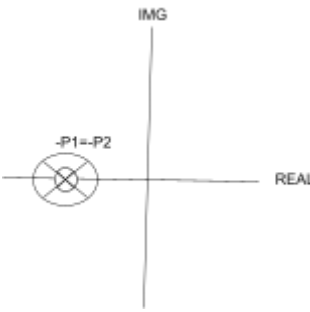
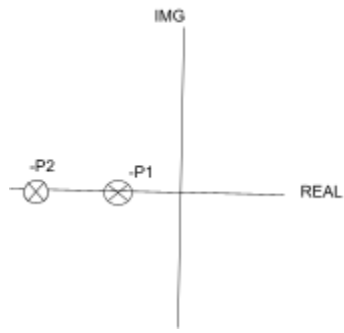
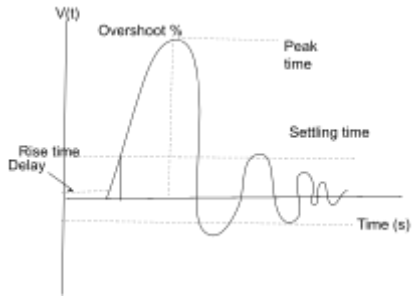
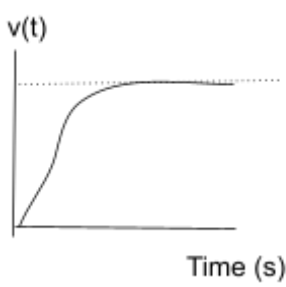
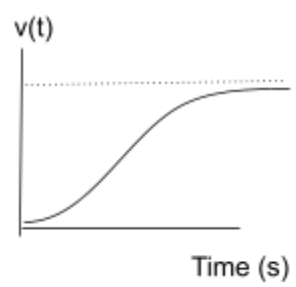
	Stable		
	Underdamped	Critically damped	Overdamped
	$0 < \xi < 1$	$\xi = 1$	$\xi > 1$
Poles	Two negative, complex poles $P_1 = -a + jb$ $P_2 = -a - jb$	Two identical, negative, real poles	Two negative, different, real poles
Stability			
Graph			
Behavior	Oscillating	Fast rise to steady state	Slower rise to steady state
Step Response	Underdamped: $V_o(t) = 1 - e^{-\beta t}(\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))$ Critically damped: $V_o(t) = 1 - [e^{-pt} + te^{-pt}]u(t)$ Over damped : $V_o(t) = [1 - (Ae^{-p_1 t} + Be^{-p_2 t})]u(t)$		

Table 2. Note that if the value of the dampening ratio was 0, the system would be unstable. Additionally, there would be a location of zeros for the stability graphs, depending on filter behavior and type.

Regarding the equations for the graph for underdamped,

- Rise time: $T_R = \frac{1+0.7\xi}{\omega_n}$
- Delay time: $T_D = \frac{0.8+2.5\xi}{\omega_n}$
- Peak time: $T_p = \frac{\pi}{\omega_d}$
- Overshoot %: $= e^{\frac{-\pi\xi}{\beta}} * 100$
- Settling time: $t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$
- $\beta = \sqrt{1 - \xi^2}$
- $\omega_n = \frac{1}{\sqrt{LC}}$
- $\omega_d = \beta\omega_n$

A second order passive circuit can also be configured into two loops. For a second order circuit consisting of two captive elements, $\frac{1}{2\pi R1C1} = Fc1$ and $\frac{1}{2\pi R2C2} = Fc2$. If the two critical frequencies are not equivalent and the high pass filter is before the low pass, the circuit exhibits band pass behavior. Since we cannot have the low pass filter before the high pass filter, a band stop filter is not possible. If the critical frequencies are the same, we have either a highpass or lowpass filter with more attenuation.

An important characteristic of filters is the -3dB cutoff frequency. The -3dB point represents the frequency at which the output power drops to one-half of the input power. The cutoff frequency occurs at the value ω where the magnitude of the transfer function decreases to $\frac{1}{\sqrt{2}}$ (or 0.707) of its maximum value. This can be found by substituting $s = j\omega$ and then finding the magnitude of the transfer function. Because the filter begins transitioning from the passband (where signals pass with minimal attenuation) to the attenuation region (where signal amplitude decreases), the -3dB point is used to define the filter's usable bandwidth. For example, for a bandpass filter the f_c of the highpass filter (located first) minus the f_c of the lowpass filter would be the bandwidth.

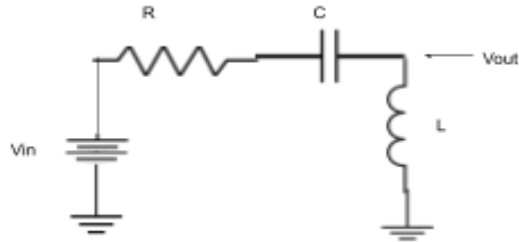
METHODOLOGY

The passive second-order circuits were built and tested to verify proper filter behavior and for analysis. The single loop circuits constructed were a second-order high-pass RLC (configured as resistor, capacitor, then inductor), a second-order low-pass RLC (configured as resistor, inductor, and then capacitor), and a second-order band pass (configured as an inductor, capacitor, and resistor). All circuits were simulated, and the highpass and bandpass circuits were built on a breadboard. The second-order circuit (a high-pass filter in combination with a low-pass) was simulated and built on a breadboard as well.

Each circuit was first designed theoretically by determining the required component values based on a desired cutoff frequency. Then, with the chosen components, the circuits were built on Multisim and the Bode plots generated. The Bode plots were used to check the -3dB cutoff frequency and verify the predicted filter behavior. In each section for each circuit, transfer functions were derived for each circuit, and analysis was done to determine filter behavior based on these transfer functions. Additionally, in each section, the simulated and calculated results were compared to verify the transfer functions and confirm the expected filtering characteristics of each circuit.

HIGH PASS FILTER:

We first need to determine the component values to use in the simulation and in the physical circuit.



$$V_{out} = \frac{LS}{LS + (R + \frac{1}{CS})} V_{in}, H(s) = \frac{V_{out}}{V_{in}} = \frac{LS}{LS + (R + \frac{1}{CS})} = \frac{LSCS}{LSCS + RCS + 1} = \frac{LCS^2}{LCS^2 + RCS + 1}$$

Normalize by dividing by the term in front of the highest power (LC)

$$H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

We can compare to the standard second order equation to find the values of components.

$$\frac{s^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{ to } \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$\text{Therefore } \frac{R}{L} = 2\xi\omega_n \text{ and } \omega_n^2 = \frac{1}{LC}$$

$$\omega_n^2 = \frac{1}{LC}, \omega_n = \frac{\sqrt{1}}{\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

If we choose $f_o = 1.59kHz$, $\omega_n = \omega_o = 2\pi f_o = 2\pi(1.59k)hz = 9.99kHz = 10kHz$ and choose $C = 0.1\mu F$

$$10k = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L(0.1\mu F)}}, L = 100mH$$

$$\frac{R}{L} = 2\xi\omega_n, R = 2L\omega_n\xi = 2[100mH][10kHz]\xi = 2000\xi$$

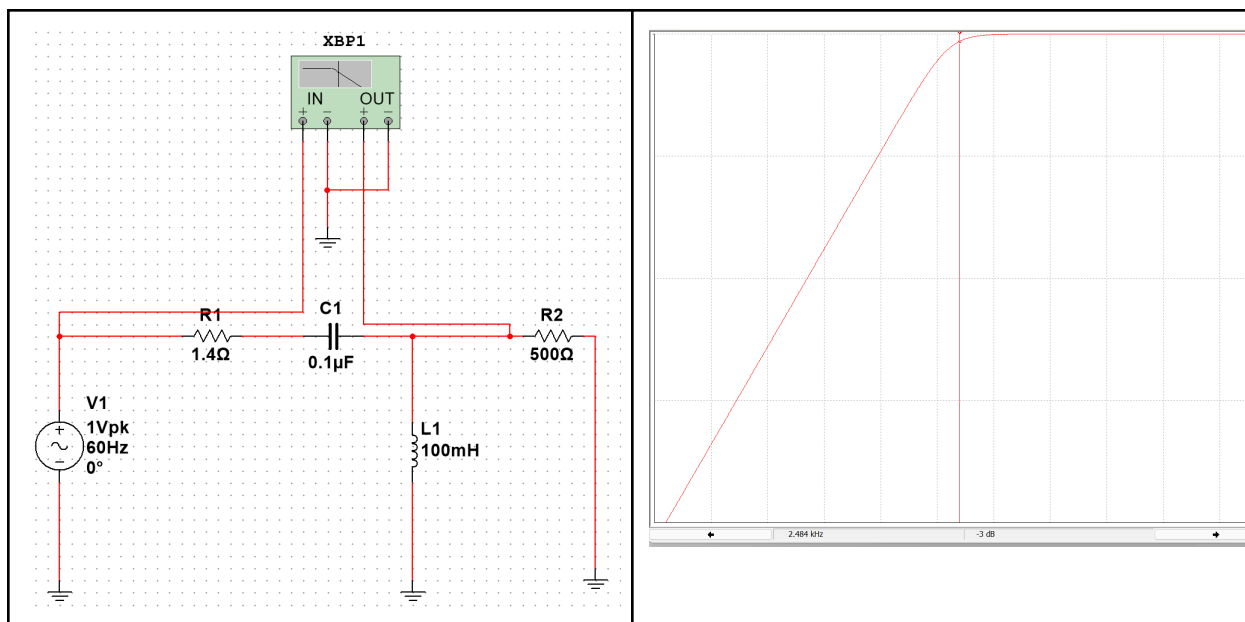
Stable		
Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$
ξ is chosen to be 0.7		ξ is chosen to be 2.2
$R = 2000\xi = 2000(0.7) = 1400\Omega$	$R = 2000\xi = 2000(1) = 2000\Omega$	$R = 2000\xi = 2000(2.2) = 4400\Omega$

Table 3.

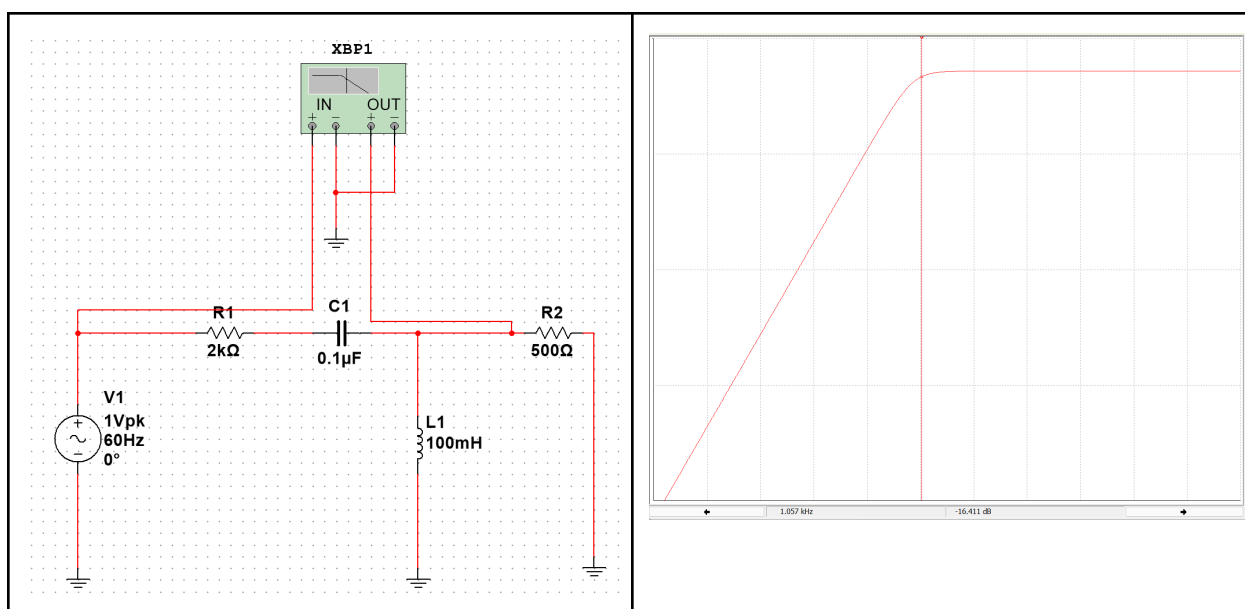
All of the chosen components are shown in Table 4.

	Circuit from Multisim	Circuit from breadboard
L	100mH	100mH
C	0.1uF	0.1uF
R (underdamped)	1,400 Ω	240 Ω
R (overdamped)	2,000 Ω	876 Ω
R (critically damped)	4,400 Ω	400 Ω

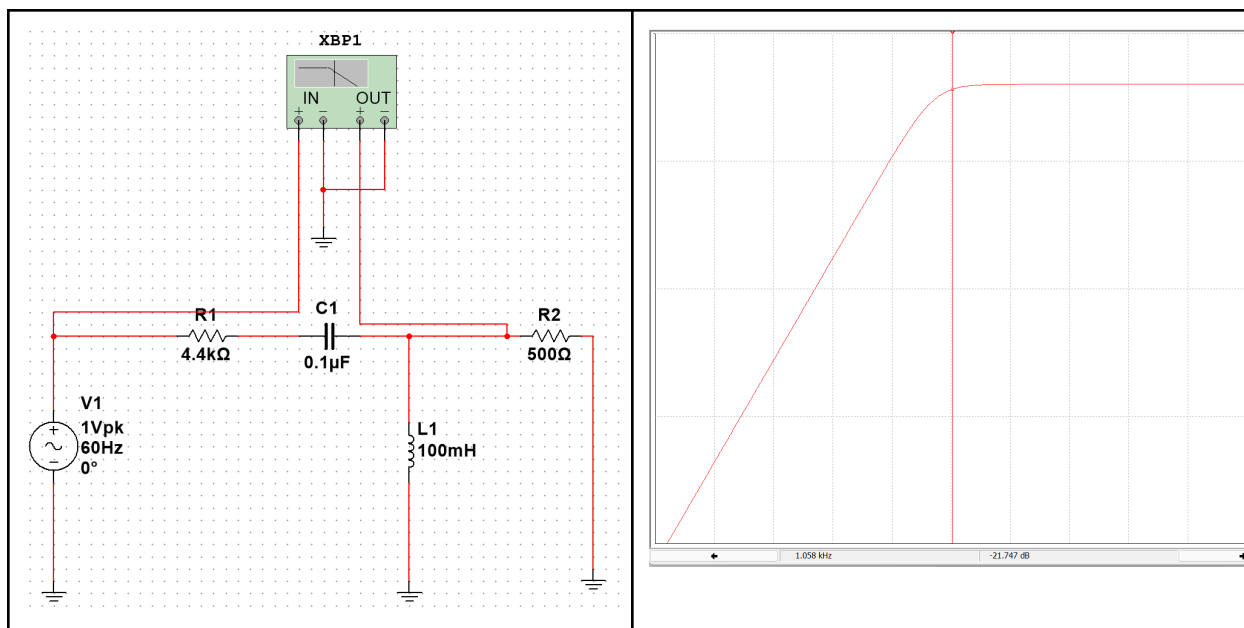
Table 4. Note that due to miscalculation errors, it was thought that $R = 400\xi$ instead of $R = 2000\xi$, and the physical circuit values were built reflecting the incorrect formula.



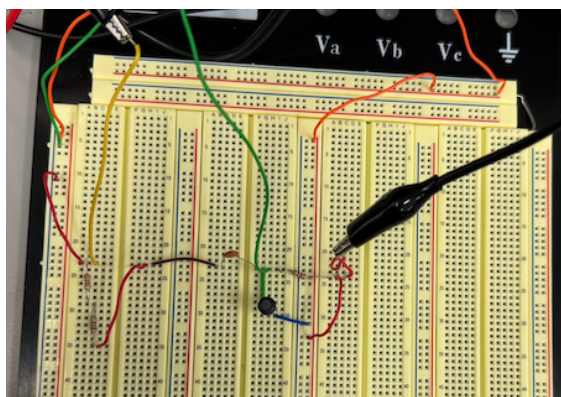
IMG 1. Bode plot of the simulated circuit, for underdamped



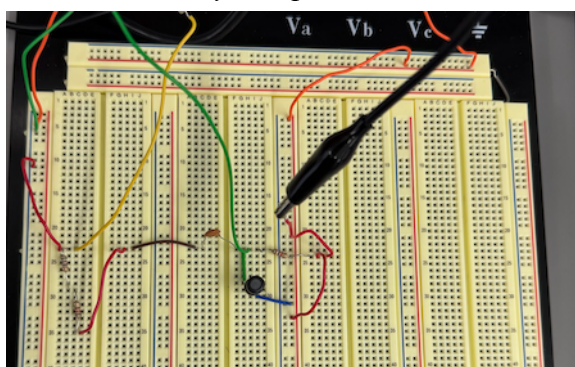
IMG 2. Bode plot of the simulated circuit, for overdamped



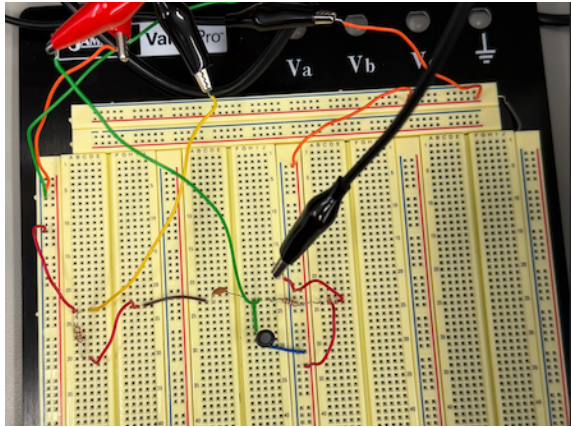
IMG 3. Bode plot of the simulated circuit, for critically damped



IMG 4. Critically dampened circuit, with a resistance of 400Ω



IMG 5. Overdamped circuit, with a resistance of 876Ω



IMG 6. Underdamped circuit, with a resistance of 220Ω

Calculations and analysis:

1. Filter behavior

Note that we have three different resistance values, but the behavior of the circuit stays the same, and we can simply leave the variable $\frac{R}{L}$ in place and not plug in a specific resistance value.

$$H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{(100mH)(0.1\mu F)}} = \frac{s^2}{s^2 + \frac{R}{L}s + 10^8}$$

$$\text{As } S \rightarrow 0, H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + 10^8} \approx \frac{0^2}{0^2 + \frac{R}{L}(0) + 10^8} \approx \frac{0}{10^8} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + 10^8} \approx \frac{\infty^2}{\infty^2 + \frac{R}{L}(\infty) + 10^8}$$

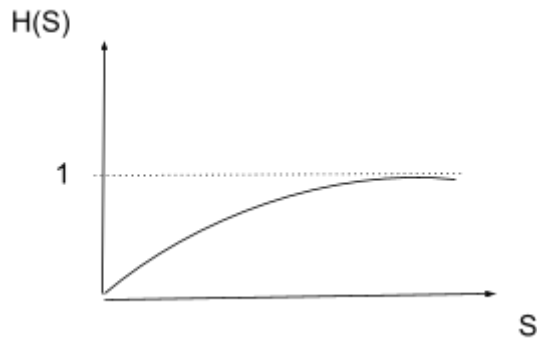
We want to use L'Hopital's rule to account for $\frac{\infty}{\infty}$, but to make it conceptually easier, we can

plug any of the three resistance values for $\frac{R}{L}$ to make it clearly a constant. $\frac{R}{L} = \frac{280}{100mH} = 2800$

$$\text{Now, } \frac{\frac{d}{ds}(s^2)}{\frac{d}{ds}(s^2 + 2800s + 10^8)} = \frac{2s^{2-1}}{2s^{2-1} + 2800(1s^{1-1}) + 0} = \frac{2s}{2s + 2800}$$

$$\text{As } S \rightarrow \infty, \frac{2s}{2s + 2800} = \frac{\infty}{\infty + 2800}$$

Using L'Hopital's rule to account for $\frac{\infty}{\infty}, \frac{\frac{d}{ds}(2s)}{\frac{d}{ds}(2s+2800)} = \frac{2(1)s^{1-1}}{2(1)s^{1-1}+0} = \frac{2}{2} = 1$, so $S \rightarrow \infty, H = 1$



Based on this analysis of the limits of s , the circuit passes high-frequency signals, thus **behaving like a high-pass filter**.

2. Stability and filter behavior analysis:

2a. Underdamped:

$$H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{s^2}{s^2 + \frac{1400}{100mH}s + \frac{1}{(100mH)(0.1\mu F)}} = \frac{s^2}{s^2 + 14,000s + 10^8} = \frac{n(Z)}{n(P)}$$

$$n(Z) = s^2, \text{ if } n(Z) = 0, s^2 = 0, s = 0 \text{ and } s = 0$$

We have a double zero at zero, indicating high-pass filter behavior.

$$n(p) = s^2 + 14,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 14,000$$

$$c = 10^8$$

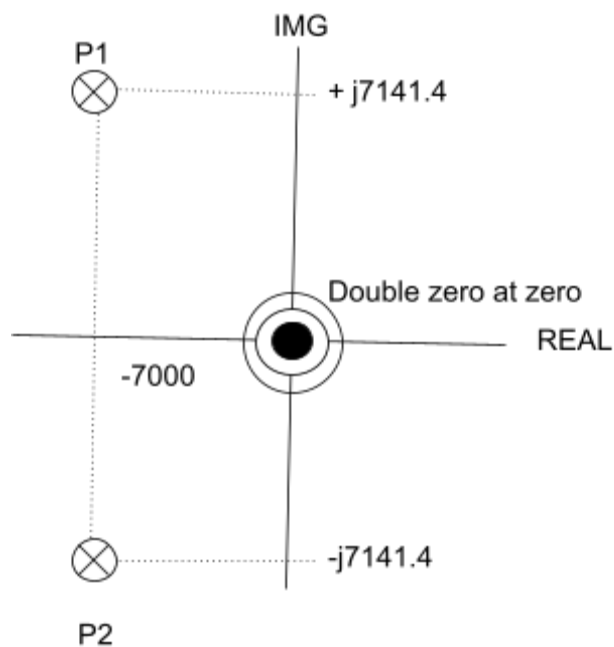
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(14,000) \pm \sqrt{(14,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -7000 \pm \frac{\sqrt{-204 \cdot 10^8}}{2} \rightarrow$$

$$-7000 \pm j7141.4$$

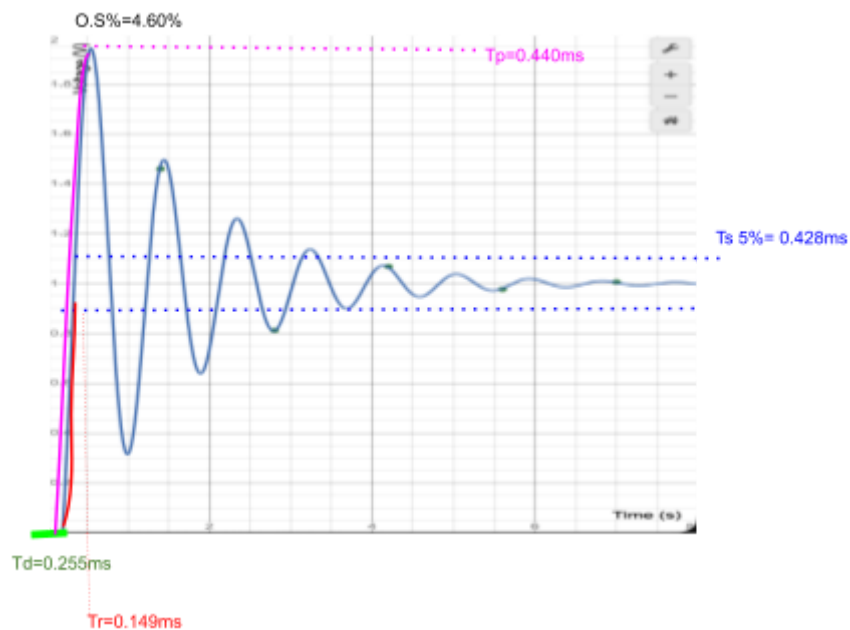
Two negative, complex poles. We have two negative poles, and the system is stable.

$$P_1 = -7000 + j7141.4$$

$$P_2 = -7000 - j7141.4$$



IMG 7. Oscillating behavior is seen



IMG 8. Please note that this image is a rough estimation and is based upon image 9.

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - 0.7^2}$	$\beta = 0.7141$
	$\omega_n = \frac{1}{\sqrt{LC}}$	$\omega_n = \frac{1}{\sqrt{(0.1H)(1 \cdot 10^{-7}F)}}$	$\omega_n = 10,000 \text{ rad/s}$

	$\omega_d = \beta\omega_n$	$\omega_d = (0.7141)(10,000 \text{ rad/s})$	$\omega_d = 7141 \text{ rad/s}$
Rise time	$T_R = \frac{1+0.7\xi}{\omega_n}$	$T_R = \frac{1+0.7(0.7)}{10,000}$	$T_R = 0.149\text{ms}$
Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$	$T_D = \frac{0.8+2.5(0.7)}{10,000}$	$T_D = 0.255\text{ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{7141}$	$T_p = 0.440\text{ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi 0.7}{0.7141}} * 100$	$O.S\% = 4.60\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.7)(10,000)}$	$t_s = 0.559\text{ms}$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.7)(10,000)}$	$t_s = 0.460\text{ms}$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.7)(10,000)}$	$t_s = 0.428\text{ms}$

Table 9.

The step response for an underdamped system is $V_o(t) = 1 - e^{-\beta t}(\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))$

$$V_o(t) = 1 - e^{-(0.7141)t}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

From the standard underdamped form, $e^{-\beta t}$ so $-\beta t = -\frac{t}{\tau}$ and $\tau = \frac{1}{\beta} = \frac{1}{0.7141} = 1.40$

For 5τ , $5\tau = 5(1.4004) = 7.002$

$$\text{At } t = 5\tau, e^{-\beta t} = e^{-\beta(5\tau)} = e^{-\beta(5\frac{1}{\beta})} = e^{-5}$$

$$V_o(t) = 1 - e^{-5}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

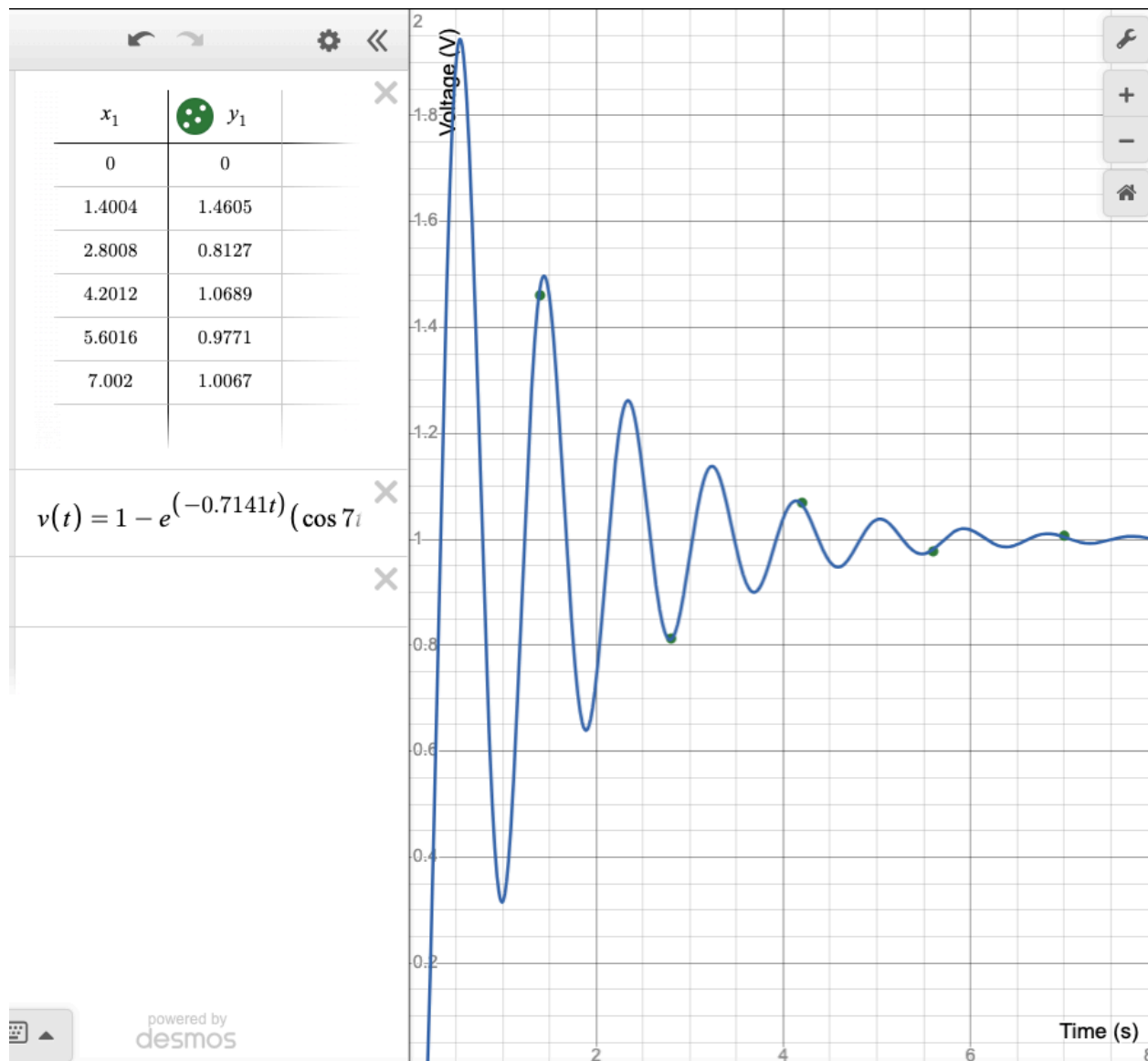
$$V_o(t) = 1 - 0.0067370(\cos(7141 * 7.002) + 0.9803 \sin(7131 * 7.002)) = 1.0067v$$

To find the % of the final value that the output is, $\frac{1.0067v}{1v} * 100 = 100.67\%$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 1.4004s$	1.461v	146.05%

$2\tau = 2.8008s$	$0.8127v$	81.27%
$3\tau = 4.2012s$	$1.0689v$	106.89%
$4\tau = 5.6016s$	$0.9771v$	97.71%
$5\tau = 7.0020s$	$1.006v$	100.67%

Table 10. $V_{out}(t)$ after time constants



IMG 9. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com/calculator). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit. The equation used in this graph is $V_o(t) = 1 - e^{-(0.7141)t}(\cos(7t) + \frac{0.7}{0.7141}\sin(7t))$ instead of $V_o(t) = 1 - e^{-(0.7141)t}(\cos(7141t) + \frac{0.7}{0.7141}\sin(7141t))$

2b. Critically dampened:

$$R = 2000\Omega, \text{ so } H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{s^2}{s^2 + \frac{2000}{100mH}s + \frac{1}{(100mH)(0.1\mu F)}} = \frac{s^2}{s^2 + 20,000s + 10^8} = \frac{n(Z)}{n(P)}$$

$$n(Z) = s^2, \text{ if } n(Z) = 0, s^2 = 0, s = 0 \text{ and } s = 0$$

We have a double zero at zero, indicating we still have high-pass filter behavior.

$$n(p) = s^2 + 20,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

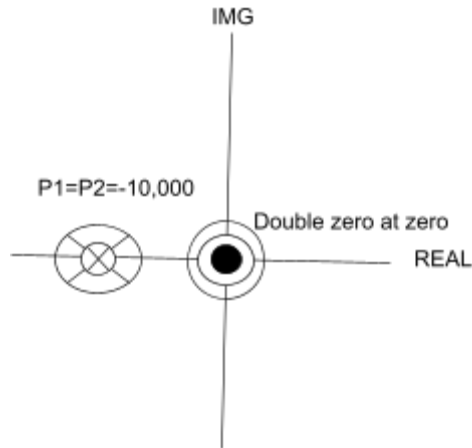
$$b = 20,000$$

$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(20,000) \pm \sqrt{(20,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -10,000 \pm \frac{\sqrt{40,000 - 40,000}}{2} \rightarrow -10,000 \pm \frac{\sqrt{0}}{2}$$

$$P_1 = P_2 = -10,000$$

Two identical, negative, real poles. We have two negative poles, and the system is stable.



IMG 10.

The step response for an underdamped system is $V_o(t) = 1 - [e^{-pt} + te^{-pt}]u(t)$

$$V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$$

$$e^{-pt} \text{ so } -pt = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{p} = \frac{1}{10,000} = 0.0001s = 0.1ms$$

$$\text{For } 5\tau, 5\tau = 5(0.1ms) = 0.0005ms$$

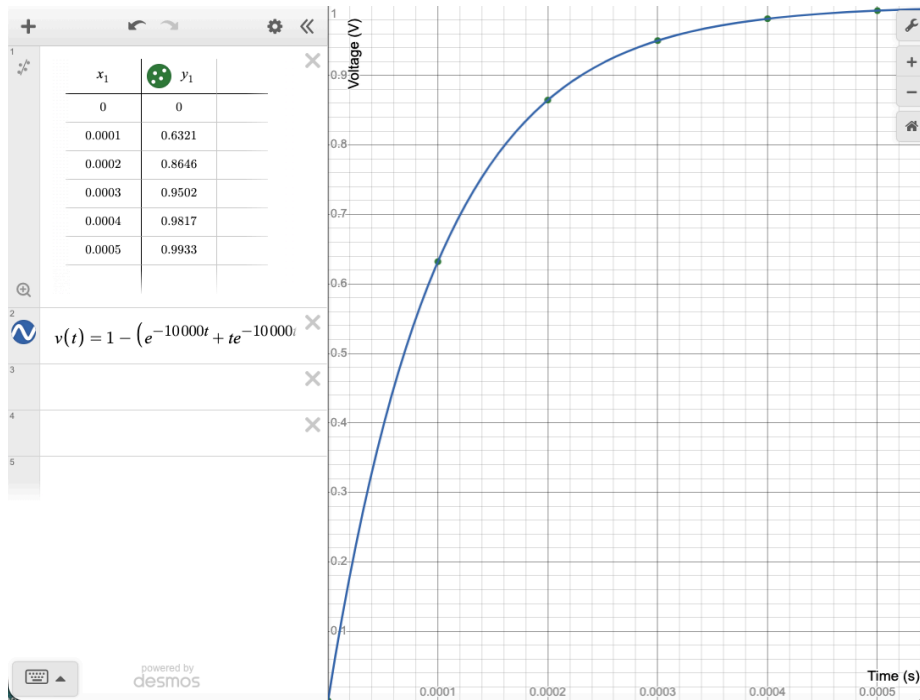
$$\text{At } t = 5\tau, e^{-pt} = e^{-p(5\tau)} = e^{-p(5\frac{1}{p})} = e^{-5}$$

$$V_o(t) = 1 - [e^{-5} + te^{-5}]u(t) = 1 - [e^{-5} + 0.0005e^{-5}]u(t) = 0.9933$$

To find the % of the final value that the output is, $\frac{0.9933v}{1v} * 100 = 99.33\%$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 0.1ms$	0.6321v	63.21%
$2\tau = 0.2ms$	0.8646v	81.27%
$3\tau = 0.3ms$	0.9502v	95.02%
$4\tau = 0.4ms$	0.9817v	98.17%
$5\tau = 0.5ms$	0.9933v	99.33%

Table 11. $V_{out}(t)$ after time constants



IMG 12. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was $V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$

2c. Overdamped

$$R = 4400\Omega, \text{ so } H(s) = \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{s^2}{s^2 + \frac{4400}{100\text{mH}}s + \frac{1}{(100\text{mH})(0.1\text{uF})}} = \frac{s^2}{s^2 + 44000s + 10^8} = \frac{n(Z)}{n(P)}$$

$$n(Z) = s^2, \text{ if } n(Z) = 0, s^2 = 0, s = 0 \text{ and } s = 0$$

We have a double zero at zero, indicating we still have high-pass filter behavior.

$$n(p) = s^2 + 40,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 40,000$$

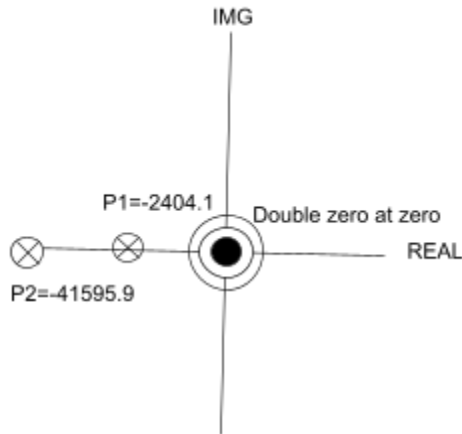
$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(40,000) \pm \sqrt{(40,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -20,000 \pm \frac{39191.8}{2}$$

$$P_1 = -2404.1$$

$$P_2 = -41595.9$$

Two different, negative, real poles. We have two negative poles, and the system is stable.



IMG 13.

The step response for an underdamped system is $V_o(t) = [1 - (Ae^{-p_1 t} + Be^{-p_2 t})]u(t)$

$$V_o(t) = [1 - (Ae^{-2404.1t} + Be^{-41595.9t})]u(t)$$

Using partial fractions: $\frac{n(s)}{(s+A)(s+B)} = \frac{A}{s+a} + \frac{B}{s+b}$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A}{(s+2404.1)} + \frac{B}{(s+41595.9)}$$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A(s+41595.9)}{(s+2404.1)(s+41595.9)} + \frac{B(s+2404.1)}{(s+2404.1)(s+41595.9)}$$

$$n(s) = A(s + 41595.9) + B(s + 2404.1)$$

Expand: $n(s) = As + A41595.9 + Bs + B2404.1$

Regroup: $n(s) = (A + B)s + A41595.9 + B2404.1$

$$(A + B)s = (1)s, A + B = 1, B = 1 - A$$

$$A41595.9 + B2404.1 = 0, A41595.9 + (1 - A)2404.1 = 0, (41595.9 - 2404.1)A + 2404.1 = 0$$

$$39191.8A + 2404.1 = 0, A = -0.0614 \text{ and } B = 1.0614.$$

Therefore, the step response is $V_o(t) = [1 - (1.0614e^{-2404.1t} + -0.0614e^{-41595.9t})]u(t)$

We need to decide which exponent to use. There are two exponentials, so technically two time

constants. I choose to use the smaller pole magnitude because it decays slowly and controls the

overall behavior more.

$$e^{-2404.1t} \text{ so } -2404t = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{2404.1} = 0.4160 \text{ ms}$$

$$\text{For } 5\tau, 5\tau = 5(0.4160 \text{ ms}) = 2.07980 \text{ ms}$$

$$\text{At } t = 5\tau, e^{-2404.1t} = e^{-2404.1(5\tau)} = e^{-2404.1(5 \cdot \frac{1}{2404.1})} = e^{-5}$$

$$V_o(t) = [1 - (1.0614e^{-5} + -0.0614e^{-41595.9(2.07980 \text{ ms})})]u(t) = 0.992848$$

$$\text{To find the \% of the final value that the output is, } \frac{0.9928 \text{ v}}{1 \text{ v}} * 100 = 99.28\%$$

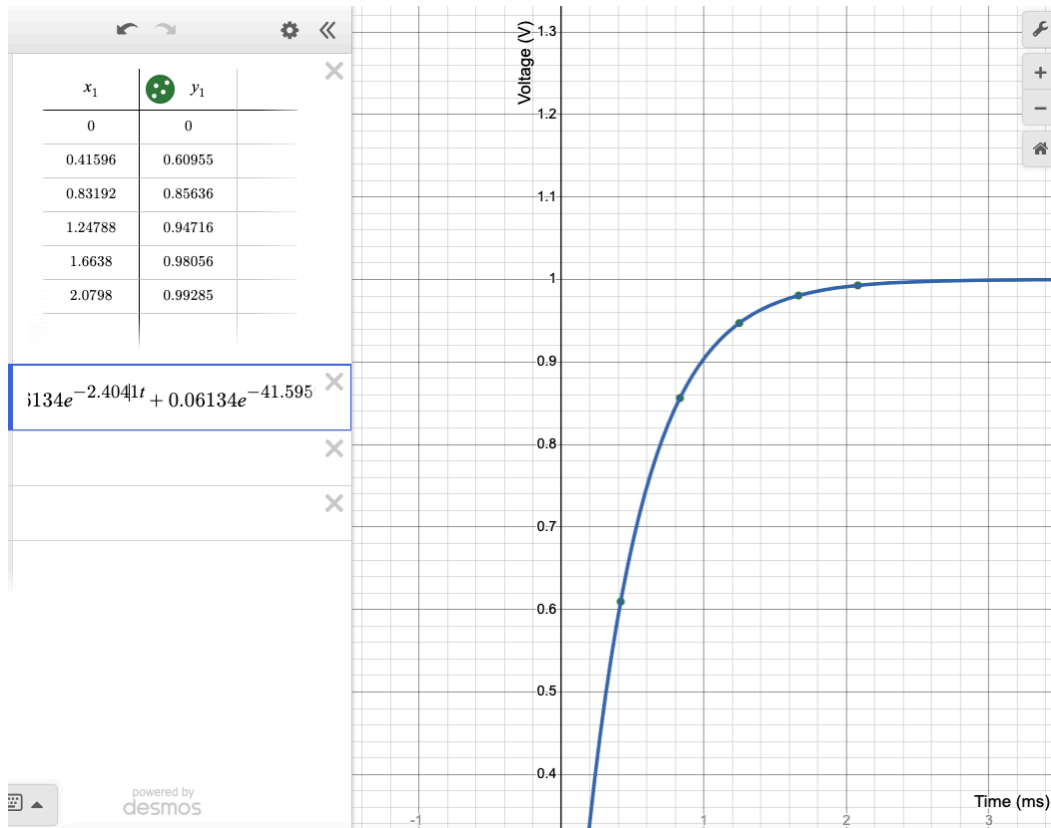
Recall that for critically dampened, at 5τ (0.5 ms) there was 0.9933 v or 99.33% of the final

voltage. It tracks that for overdampened at 5τ (2.07 ms) there is slightly less voltage (0.9928 v)

and 99.28% of the final voltage, showing a slower rise time and less voltage.

	$V_{out}(t)$	% Final
0τ	0 v	0%
$1\tau = 0.416 \text{ ms}$	0.6096 v	60.96%
$2\tau = 0.8319 \text{ ms}$	0.8564 v	85.64%
$3\tau = 1.2479 \text{ ms}$	0.9471 v	94.71%
$4\tau = 1.6638 \text{ ms}$	0.9805 v	98.05%
$5\tau = 2.0798 \text{ ms}$	0.9928 v	99.28%

Table 12. $V_{out}(t)$ after time constants



IMG 14. Steady state is reached after 5τ , graphed using [Desmos.com](https://desmos.com). Note that the equation of the graph was $V_o(t) = 1 - (1.0614e^{-2.4041t} + -0.06134e^{-41.595t})$

Results:

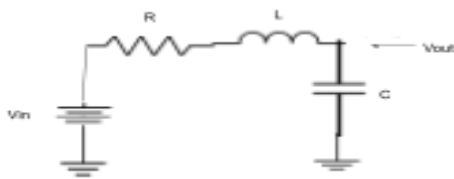
It was determined that the resistance for underdamped was 1400Ω , for critically damped 2000Ω and for over damped 4400Ω . Analysis of the transfer function using the limits of s showed that the circuit attenuates low frequencies and passes high frequencies, confirming that it behaves as a high pass filter. For the case of underdamped, it is found that there are two negative complex poles (indicating stability) and we have a double zero at zero (confirming high pass filter behavior). The oscillatory response observed in both the real time graph and the $v_{out}(t)$ versus τ is consistent with expected oscillatory behavior. Additionally, the calculated percent overshoot of 4.60% tracks well with the oscillating behavior, and all three of the settling time

calculations correspond when observing the graph in image 9. The exact points of T_R , T_D , T_p , t_s can be found in the graph in image 9 to confirm the calculations and accuracy of the formulas. Overall, the underdamped response matches the expected oscillatory behavior of a second order passive filter.

For critically damped, we see that we have two identical negative real poles, insidcating a stable system, and a double zero at zero confirming high pass filter behavior. For overdamped we have two different, negative, real poles with a double zero at zero (confirming stability and high pass filter behavior). For critically damped, at 5τ ($0.5ms$) there was $0.9933v$ or 99.33% of the final voltage. It tracks that for overdamped at 5τ ($2.07ms$) there is slightly less voltage ($0.9928v$) and 99.28% of the final voltage, showing a *slower* rise time and less voltage.

LOW PASS FILTER

We first need to determine the component values to use in the simulation and in the physical circuit.



$$V_{out} = \frac{\frac{1}{CS}}{\frac{1}{CS} + (R + LS)} V_{in}, H(s) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + RCS + LCS^2} = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}, H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$$

We can compare to the standard second order equation to find the values of components.

$$\frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{ to } \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$$

Therefore $\frac{R}{L} = 2\xi\omega_n$ and $\omega_n^2 = \frac{1}{LC}$

$$\omega_n^2 = \frac{1}{LC}, \quad \omega_n = \frac{\sqrt{1}}{\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

If we choose $f_o = 1.59kHz$, $\omega_n = \omega_o = 2\pi f_o = 2\pi(1.59k)hz = 9.99khz = 10khz$ and choose $C = 0.1\mu F$

$$10k = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L(0.1\mu F)}}, \quad L = 100mH$$

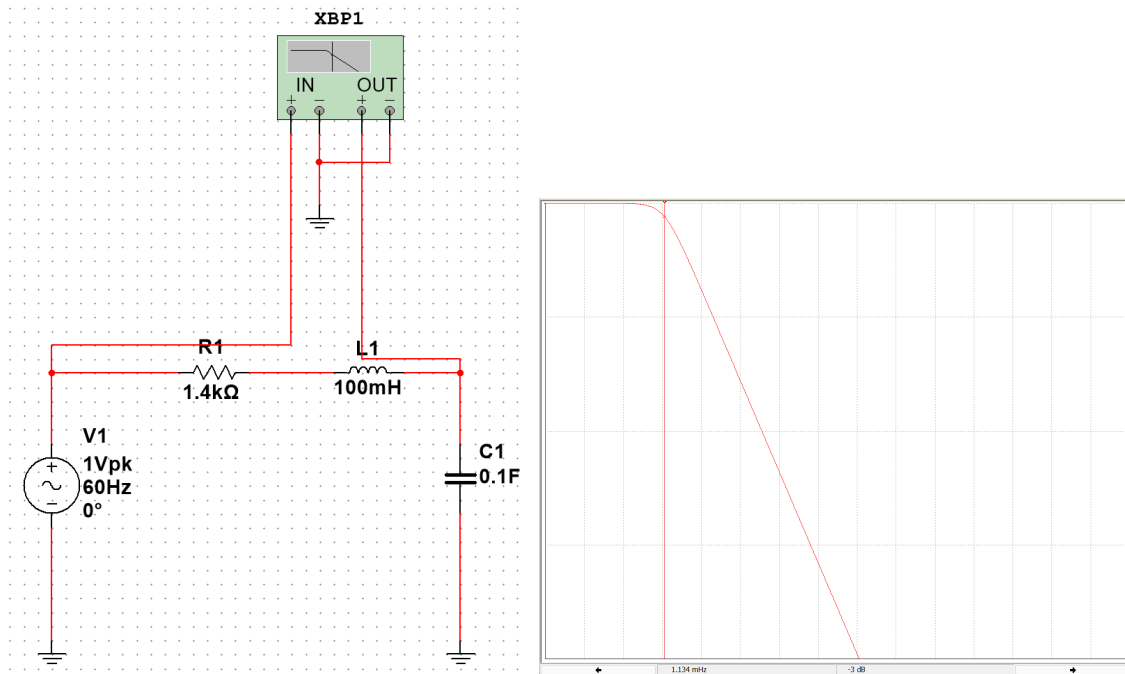
$$\frac{R}{L} = 2\xi\omega_n, \quad R = 2L\omega_n\xi = 2[100mH][10khz]\xi = 2000\xi$$

Stable		
Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$
ξ is chosen to be 0.7		ξ is chosen to be 2.2
$R = 2000\xi = 2000(0.7) = 1400\Omega$	$R = 2000\xi = 2000(1) = 2000\Omega$	$R = 2000\xi = 2000(2.2) = 4400\Omega$

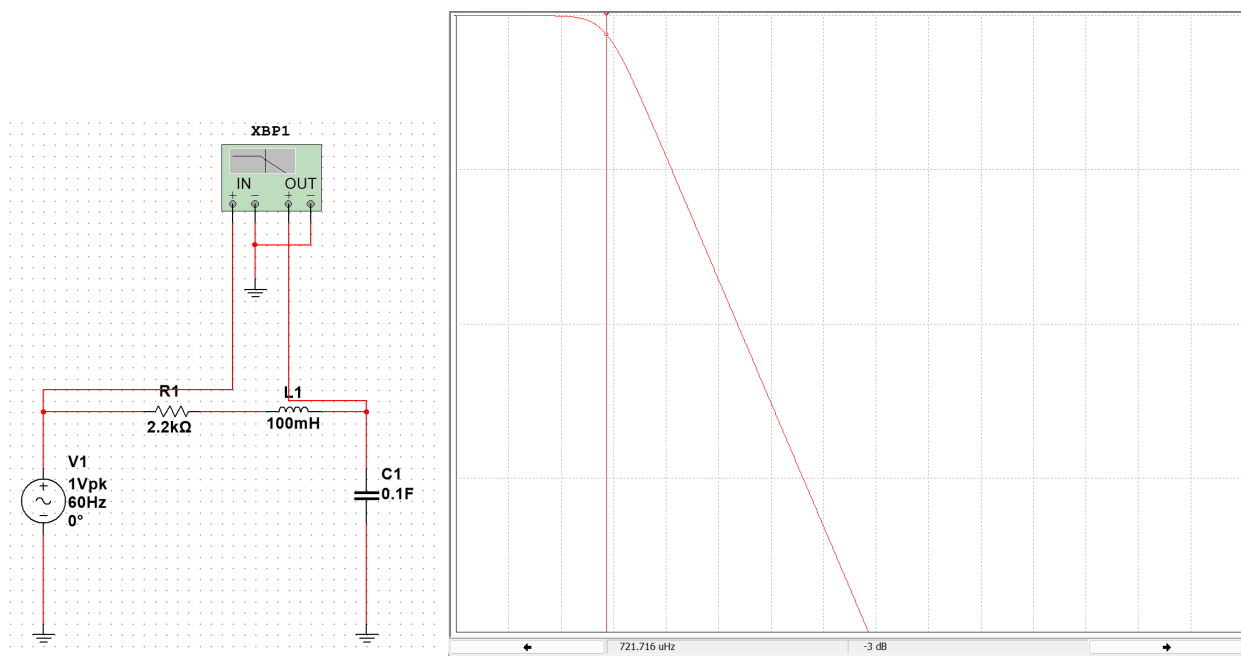
Table 13.

	Circuit from Multisim
L	100mH
C	0.1F
R (underdamped)	1,400 Ω
R (overdamped)	2,000 Ω
R (critically damped)	4,400 Ω

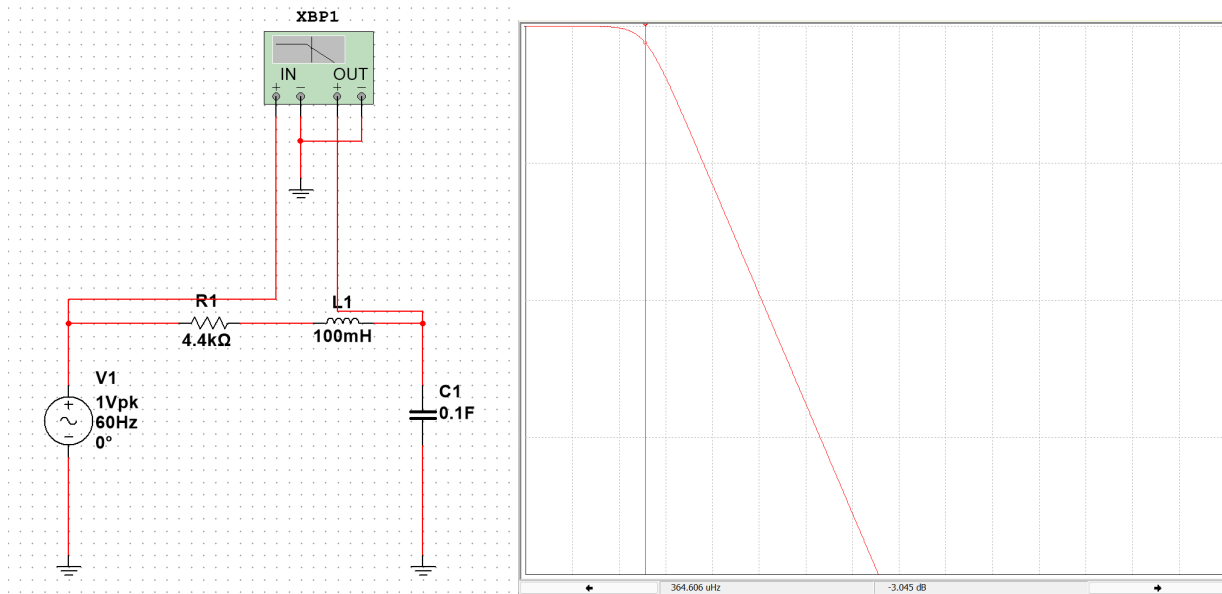
Table 14



IMG 15. Bode plot of the simulated circuit, for underdamped



IMG 16. Bode plot of the simulated circuit, for overdamped



IMG 17. Bode plot of the simulated circuit, for critically damped

Calculations and analysis:

1. Filter behavior

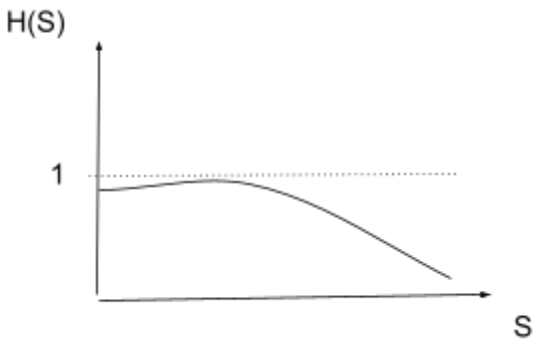
Note that we have three different resistance values, but the behavior of the circuit stays the same

and we can simply leave the variable $\frac{R}{L}$ in place and not plug in a specific resistance value.

$$H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})} = \frac{\frac{1}{(100mH)(0.1uF)}}{s^2 + \frac{R}{L}s + \frac{1}{(100mH)(0.1uF)}} = \frac{10^8}{s^2 + (\frac{R}{L})s + (10^8)}$$

$$\text{As } S \rightarrow 0, H(s) = \frac{10^8}{s^2 + (\frac{R}{L})s + (10^8)} \approx \frac{10^8}{0^2 + (\frac{R}{L})0 + (10^8)} \approx \frac{10^8}{10^8} \approx 1$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{10^8}{s^2 + (\frac{R}{L})s + (10^8)} \approx \frac{10^8}{\infty^2 + (\frac{R}{L})\infty + (10^8)} \approx \frac{10^8}{\infty} \approx 0$$



Based on this analysis of the limits of s , the circuit passes high-frequency signals, thus **behaving like a low-pass filter**.

2. Stability and filter behavior analysis:

2a. Underdamped:

$$H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})} = \frac{\frac{1}{(100mH)(0.1uF)}}{s^2 + \frac{1400}{100mH}s + \frac{1}{(100mH)(0.1uF)}} = \frac{10^8}{s^2 + (14,000)s + (10^8)} = \frac{n(Z)}{n(P)}$$

$n(Z) = 10^8$, if $n(Z) = 0$, $10^8 = 0$

Since the numerator is a constant, there are no finite zeros, and thus we have a zero at infinity, which indicates low pass filter behavior.

$$n(p) = s^2 + 14,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 14,000$$

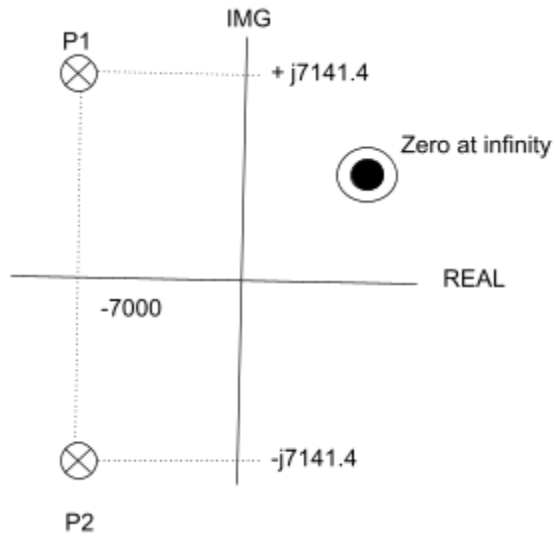
$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(14,000) \pm \sqrt{(14,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -7000 \pm \frac{\sqrt{-204 \cdot 10^8}}{2} \rightarrow -7000 \pm j7141.4$$

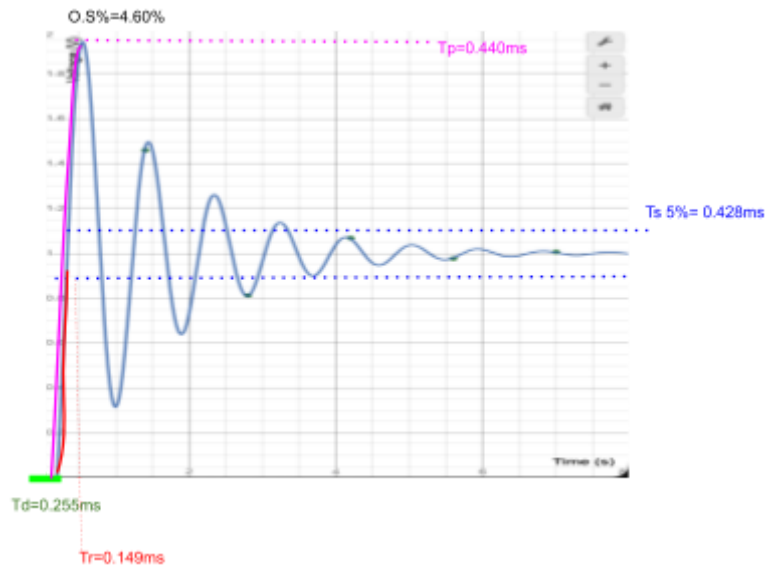
Two negative, complex poles. We have two negative poles, and the system is stable.

$$P_1 = -7000 + j7141.4$$

$$P_2 = -7000 - j7141.4$$



IMG 18. Oscillating behavior is seen



IMG 19. Please note that this image is a rough estimation and is based on image 9.

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - 0.7^2}$	$\beta = 0.7141$
	$\omega_n = \frac{1}{\sqrt{LC}}$	$\omega_n = \frac{1}{\sqrt{(0.1H)(1 \cdot 10^{-7}F)}}$	$\omega_n = 10,000 \text{ rad/s}$
	$\omega_d = \beta \omega_n$	$\omega_d = (0.7141)(10,000 \text{ rad/s})$	$\omega_d = 7141 \text{ rad/s}$
Rise time	$T_R = \frac{1 + 0.7\xi}{\omega_n}$	$T_R = \frac{1 + 0.7(0.7)}{10,000}$	$T_R = 0.149 \text{ ms}$

Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$	$T_D = \frac{0.8+2.5(0.7)}{10,000}$	$T_D = 0.255\text{ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{7141}$	$T_p = 0.440\text{ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi 0.7}{0.7141}} * 100$	$O.S\% = 4.60\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.7)(10,000)}$	$t_s = 0.559\text{ms}$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.7)(10,000)}$	$t_s = 0.460\text{ms}$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.7)(10,000)}$	$t_s = 0.428\text{ms}$

Table 15.

The step response for an underdamped system is $V_o(t) = 1 - e^{-\beta t}(\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))$

$$V_o(t) = 1 - e^{-(0.7141)t}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

From the standard underdamped form, $e^{-\beta t}$ so $-\beta t = -\frac{t}{\tau}$ and $\tau = \frac{1}{\beta} = \frac{1}{0.7141} = 1.40$

For 5τ , $5\tau = 5(1.4004) = 7.002$

At $t = 5\tau$, $e^{-\beta t} = e^{-\beta(5\tau)} = e^{-\beta(5\frac{1}{\beta})} = e^{-5}$

$$V_o(t) = 1 - e^{-5}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

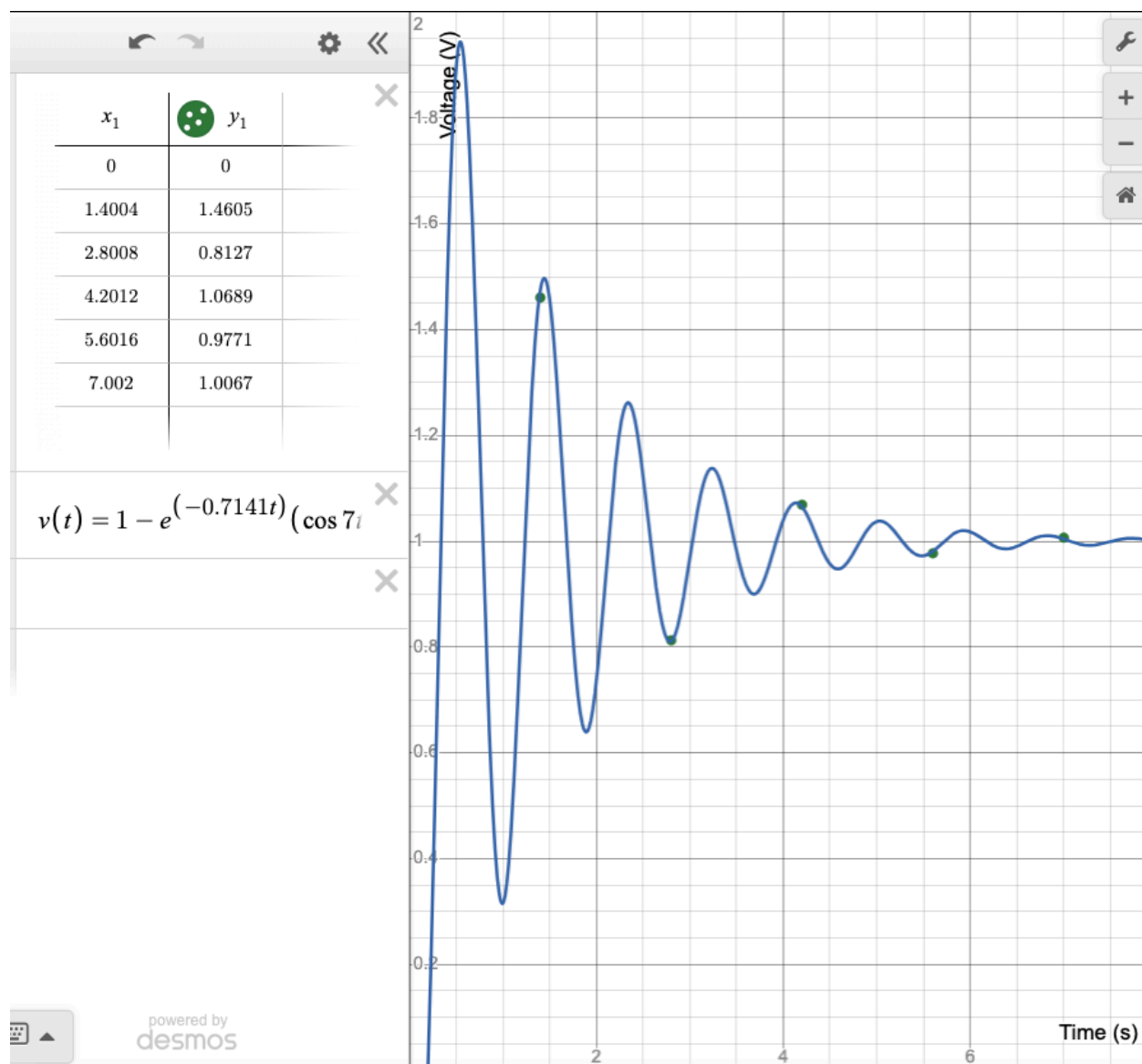
$$V_o(t) = 1 - 0.0067370(\cos(7141 * 7.002) + 0.9803 \sin(7131 * 7.002)) = 1.0067v$$

To find the % of the final value that the output is, $\frac{1.0067v}{1v} * 100 = 100.67\%$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 1.4004s$	1.461v	146.05%
$2\tau = 2.8008s$	0.8127v	81.27%
$3\tau = 4.2012s$	1.0689v	106.89%
$4\tau = 5.6016s$	0.9771v	97.71%

$5\tau = 7.0020s$	1.006v	100.67%
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Table 16. $V_{out}(t)$ after time constants



IMG 20. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line

of best fit. The equation used in this graph is $V_o(t) = 1 - e^{-(0.7141)t}(\cos(7t) + \frac{0.7}{0.7141}\sin(7t))$

instead of $V_o(t) = 1 - e^{-(0.7141)t}(\cos(7141t) + \frac{0.7}{0.7141}\sin(7141t))$

2b. Critically dampened:

$$R = 2000\Omega, \text{ so } H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})} = \frac{\frac{1}{(100mH)(0.1\mu F)}}{s^2 + \frac{2000}{100mH}s + \frac{1}{(100mH)(0.1\mu F)}} = \frac{10^8}{s^2 + (20,000)s + (10^8)} = \frac{n(Z)}{n(P)}$$

$n(Z) = 10^8$, if $n(Z) = 0$, $10^8 = 0$

Since the numerator is a constant, there are no finite zeros, and thus we have a zero at infinity, which indicates low pass filter behavior.

$$n(p) = s^2 + 20,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

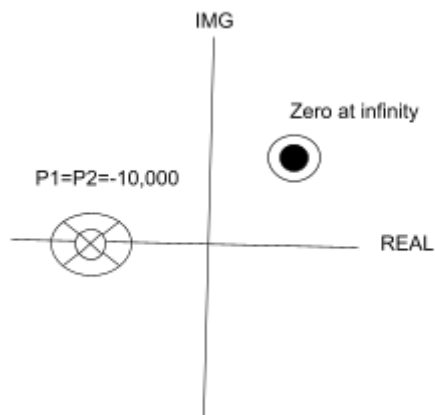
$$b = 20,000$$

$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(20,000) \pm \sqrt{(20,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -10,000 \pm \frac{\sqrt{40,000 - 40,000}}{2} \rightarrow -10,000 \pm \frac{\sqrt{0}}{2}$$

$$P_1 = P_2 = -10,000$$

Two identical, negative, real poles. We have two negative poles, and the system is stable.



IMG 21.

The step response for an underdamped system is $V_o(t) = 1 - [e^{-pt} + te^{-pt}]u(t)$

$$V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$$

$$e^{-pt} \text{ so } -pt = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{p} = \frac{1}{10,000} = 0.0001s = 0.1ms$$

$$\text{For } 5\tau, 5\tau = 5(0.1ms) = 0.0005ms$$

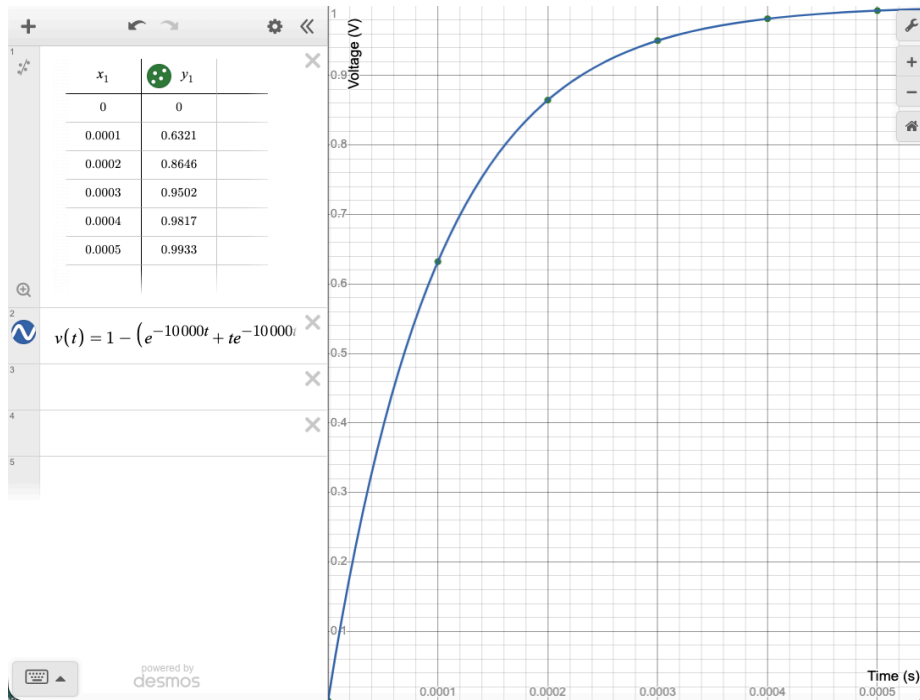
$$\text{At } t = 5\tau, e^{-pt} = e^{-p(5\tau)} = e^{-p(5\frac{1}{p})} = e^{-5}$$

$$V_o(t) = 1 - [e^{-5} + te^{-5}]u(t) = 1 - [e^{-5} + 0.0005e^{-5}]u(t) = 0.9933$$

To find the % of the final value that the output is, $\frac{0.9933v}{1v} * 100 = 99.33\%$

	$V_{out}(t)$	% Final
0τ	$0v$	0%
$1\tau = 0.1ms$	$0.6321v$	63.21%
$2\tau = 0.2ms$	$0.8646v$	81.27%
$3\tau = 0.3ms$	$0.9502v$	95.02%
$4\tau = 0.4ms$	$0.9817v$	98.17%
$5\tau = 0.5ms$	$0.9933v$	99.33%

Table 17. $V_{out}(t)$ after time constants



IMG 22. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was $V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$

2c. Overdamped

$$R = 4400\Omega, \text{ so } H(s) = \frac{\frac{1}{LC}}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})} = \frac{\frac{1}{(100mH)(0.1\mu F)}}{s^2 + \frac{4400}{100mH}s + \frac{1}{(100mH)(0.1\mu F)}} = \frac{10^8}{s^2 + (44,000)s + (10^8)} = \frac{n(Z)}{n(P)}$$

$n(Z) = 10^8$, if $n(Z) = 0$, $10^8 = 0$

Since the numerator is a constant, there are no finite zeros, and thus we have a zero at infinity, which indicates low-pass filter behavior.

$$n(Z) = s^2, \text{ if } n(Z) = 0, s^2 = 0, s = 0 \text{ and } s = 0$$

We have a double zero at zero, indicating we still have high-pass filter behavior.

$$n(p) = s^2 + 40,000s + 10^8$$

To set, $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 40,000$$

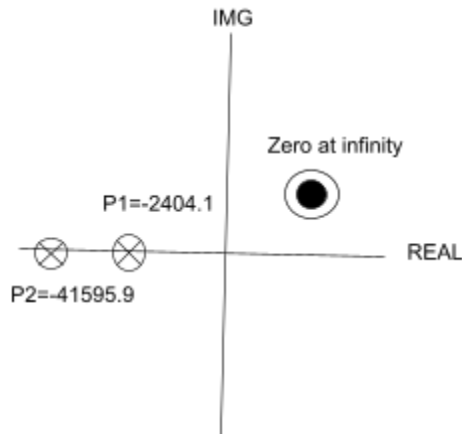
$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(40,000) \pm \sqrt{(40,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -20,000 \pm \frac{39191.8}{2}$$

$$P_1 = -2404.1$$

$$P_2 = -41595.9$$

Two different, negative, real poles. We have two negative poles, and the system is stable.



IMG 23.

The step response for an underdamped system is $V_o(t) = [1 - (Ae^{-p_1 t} + Be^{-p_2 t})]u(t)$

$$V_o(t) = [1 - (Ae^{-2404.1t} + Be^{-41595.9t})]u(t)$$

Using partial fractions: $\frac{n(s)}{(s+A)(s+B)} = \frac{A}{s+a} + \frac{B}{s+b}$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A}{(s+2404.1)} + \frac{B}{(s+41595.9)}$$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A(s+41595.9)}{(s+2404.1)(s+41595.9)} + \frac{B(s+2404.1)}{(s+2404.1)(s+41595.9)}$$

$$n(s) = A(s + 41595.9) + B(s + 2404.1)$$

Expand: $n(s) = As + A41595.9 + Bs + B2404.1$

Regroup: $n(s) = (A + B)s + A41595.9 + B2404.1$

$$(A + B)s = (1)s, A + B = 1, B = 1 - A$$

$$A41595.9 + B2404.1 = 0, A41595.9 + (1 - A)2404.1 = 0, (41595.9 - 2404.1)A + 2404.1 = 0$$

$$39191.8A + 2404.1 = 0, A = -0.0614 \text{ and } B = 1.0614.$$

Therefore the step response is $V_o(t) = [1 - (1.0614e^{-2404.1t} + -0.0614e^{-41595.9t})]u(t)$

We need to decide which exponent to use. There's two exponentials so technically two time constants. I choose to use the smaller pole magnitude because it decays slowly and controls the overall behavior more.

$$e^{-2404.1t} \text{ so } -2404t = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{2404.1} = 0.4160 \text{ ms}$$

$$\text{For } 5\tau, 5\tau = 5(0.4160\text{ms}) = 2.07980\text{ms}$$

$$\text{At } t = 5\tau, e^{-2404.1t} = e^{-2404.1(5\tau)} = e^{-2404.1(5\frac{1}{2404.1})} = e^{-5}$$

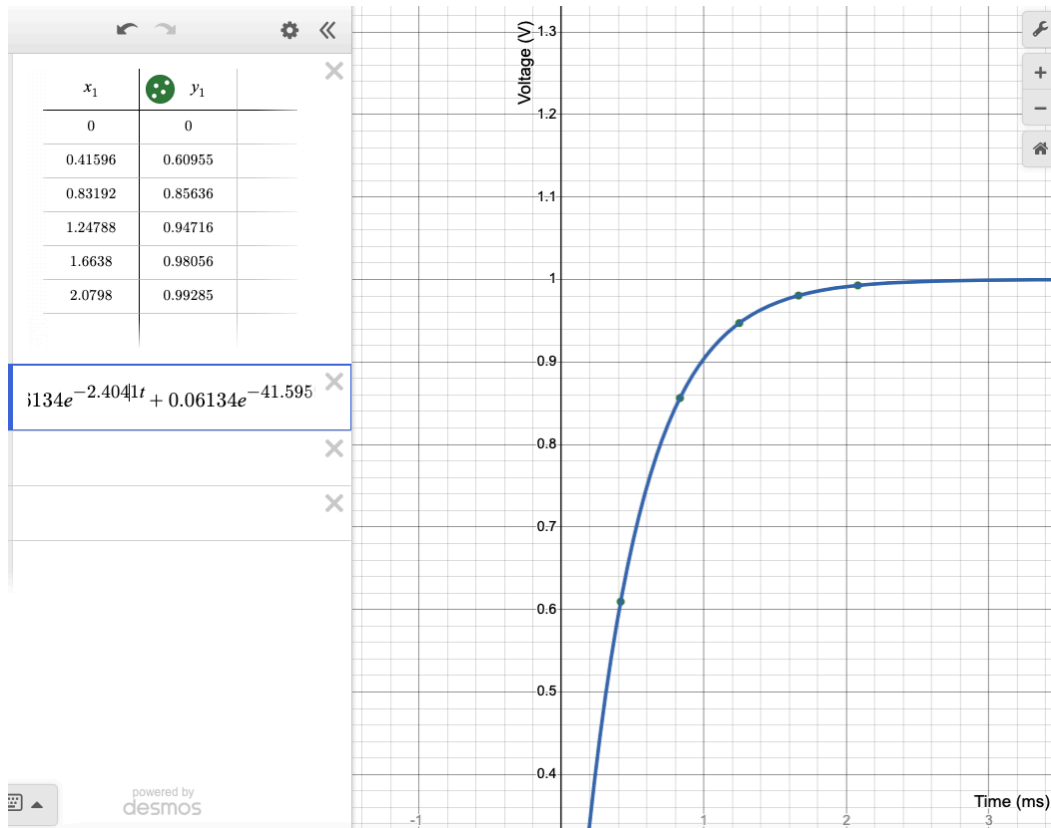
$$V_o(t) = [1 - (1.0614e^{-5} + -0.0614e^{-41595.9(2.07980\text{ms})})]u(t) = 0.992848$$

$$\text{To find the \% of the final value that the output is, } \frac{0.9928\text{v}}{1\text{v}} * 100 = 99.28\%$$

Recall that for critically dampened, at 5τ (0.5ms) there was 0.9933v or 99.33% of the final voltage. It tracks that for overdampened at 5τ (2.07ms) there is slightly less voltage (0.9928v) and 99.28% of the final voltage, showing a slower rise time and less voltage.

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 0.416\text{ms}$	0.6096v	60.96%
$2\tau = 0.8319\text{ms}$	0.8564v	85.64%
$3\tau = 1.2479\text{ms}$	0.9471v	94.71%
$4\tau = 1.6638\text{ms}$	0.9805v	98.05%
$5\tau = 2.0798\text{ms}$	0.9928v	99.28%

Table 18. $V_{out}(t)$ after time constants



IMG 24. Steady state is reached after 5τ , graphed using [Desmos.com](https://desmos.com). Note that the equation of the graph was $V_o(t) = 1 - (1.0614e^{-2.4041t} + -0.06134e^{-41.595t})$

Results:

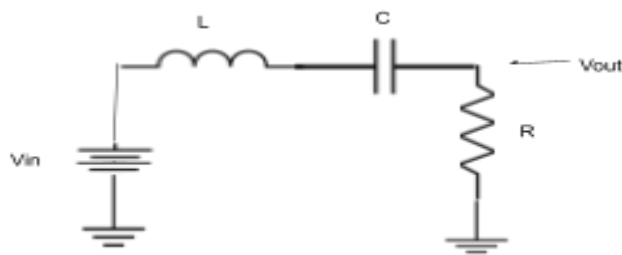
It was determined that the resistance for underdamped was 1400Ω , for critically damped 2000Ω , and for over-damped 4400Ω . These resistance values were reused from the high-pass filter calculation and analysis portion for simplification. As a result, the process and many results were quite similar. Analysis of the transfer function using the limits of s showed that the circuit attenuates high frequencies and passes low frequencies, confirming that it behaves as a low-pass filter. For the case of underdamped, it is found that there are two negative complex poles (indicating stability) and we have a zero at infinity (confirming low pass filter behavior). The oscillatory response observed in both the real time graph and the $v_{out}(t)$ versus τ

is consistent with expected oscillatory behavior. Additionally, the calculated percent overshoot of 4.60% tracks well with the oscillating behavior, and all three of the settling time calculations correspond when observing the graph in image 9. The exact points of T_R, T_D, T_p, t_s can be found in the graph in image 9 to confirm the calculations and accuracy of the formulas. Overall, the underdamped response matches the expected oscillatory behavior of a second order passive filter.

For critically damped, we see that we have two identical negative real poles, insidating a stable system, and a zero at infinity confirming low pass filter behavior. For overdamped we have two different, negative, real poles with a zero at infinity (confirming low pass filter behavior). For critically damped, at 5τ (0.5ms) there was 0.9933v or 99.33% of the final voltage. It tracks that for overdamped at 5τ (2.07ms) there is slightly less voltage (0.9928v) and 99.28% of the final voltage, showing a *slower* rise time and less voltage.

BANDPASS

We first need to determine the component values to use in the simulation and in the physical circuit.



$$V_{out} = \frac{R}{LS + \frac{1}{CS} + R} V_{in}, H(s) = \frac{V_{out}}{V_{in}} = \frac{RCS}{LCS^2 + 1 + RCS} = \frac{\frac{CS}{LC}R}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}, H(S) = \frac{\frac{R}{L}S}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$$

We can compare to the standard second order equation to find the values of components.

$$\frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{ to } \frac{\frac{R}{L}s}{s^2 + (\frac{R}{L})s + (\frac{1}{LC})}$$

$$\text{Therefore } \frac{R}{L} = 2\xi\omega_n \text{ and } \omega_n^2 = \frac{1}{LC}$$

$$\omega_n^2 = \frac{1}{LC}, \omega_n = \frac{\sqrt{1}}{\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

If we choose $f_o = 1.59\text{kHz}$, $\omega_n = \omega_o = 2\pi f_o = 2\pi(1.59\text{kHz}) = 9.99\text{kHz} = 10\text{kHz}$ and choose $C = 0.1\mu\text{F}$

$$10\text{k} = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L(0.1\mu\text{F})}}, L = 100\text{mH}$$

$$\frac{R}{L} = 2\xi\omega_n, R = 2L\omega_n\xi = 2[100\text{mH}][10\text{kHz}]\xi = 2000\xi$$

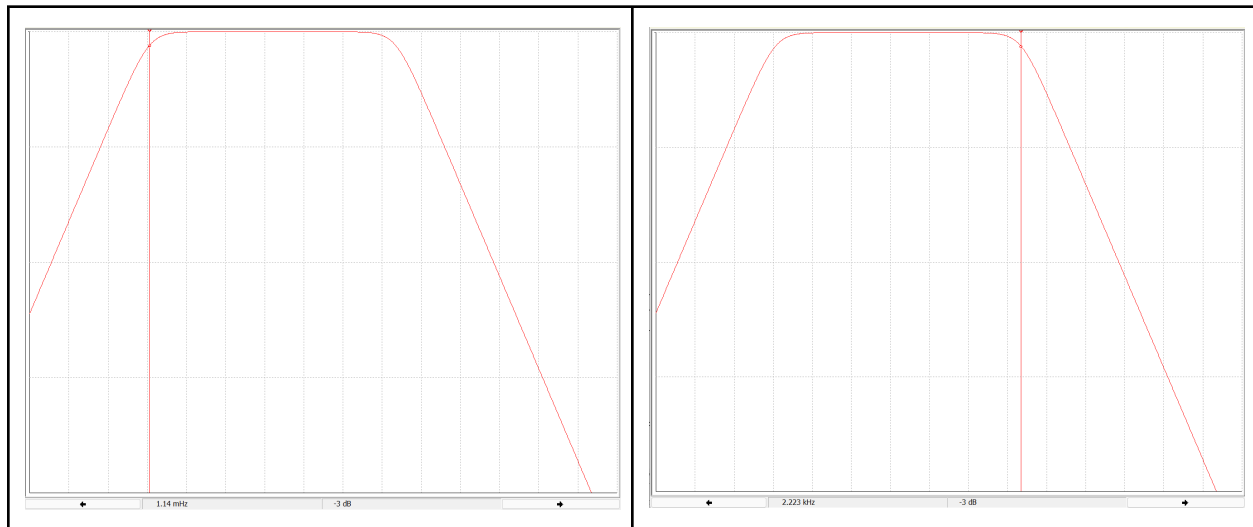
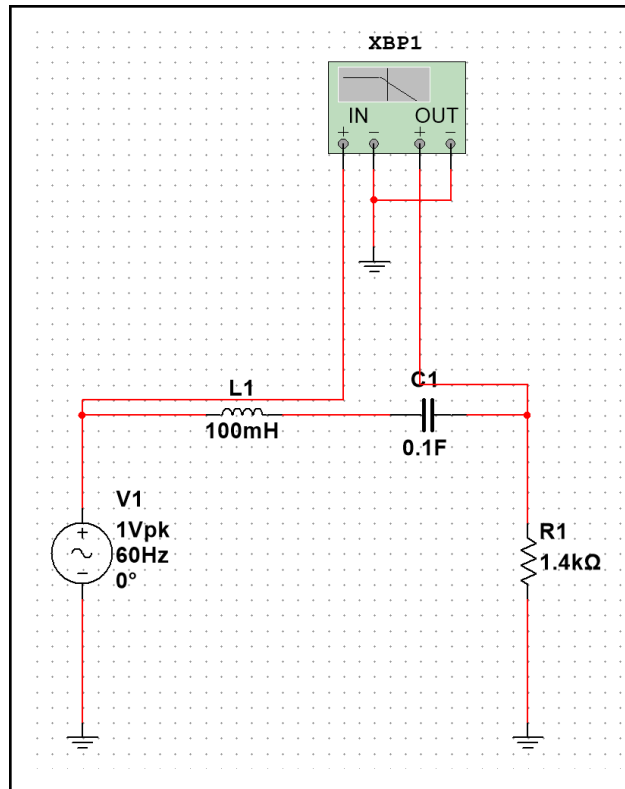
Stable		
Underdamped	Critically damped	Overdamped
$0 < \xi < 1$	$\xi = 1$	$\xi > 1$
ξ is chosen to be 0.7		ξ is chosen to be 2.2
$R = 2000\xi = 2000(0.7) = 1400\Omega$	$R = 2000\xi = 2000(1) = 2000\Omega$	$R = 2000\xi = 2000(2.2) = 4400\Omega$

Table 19. Dampening ratio and general aspects of the circuit remain the same compared to LP

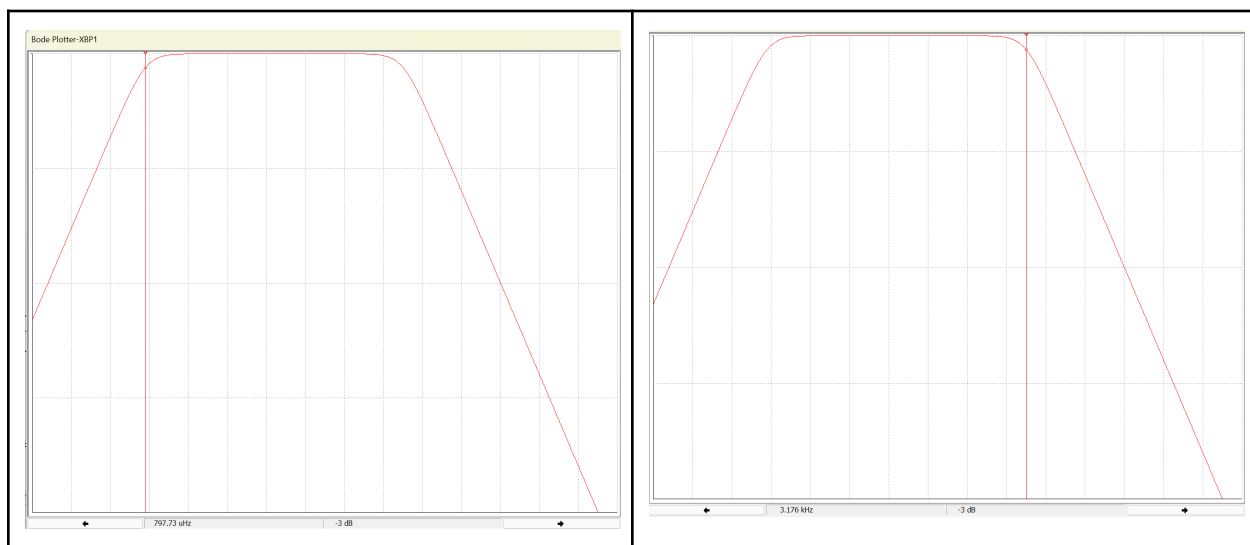
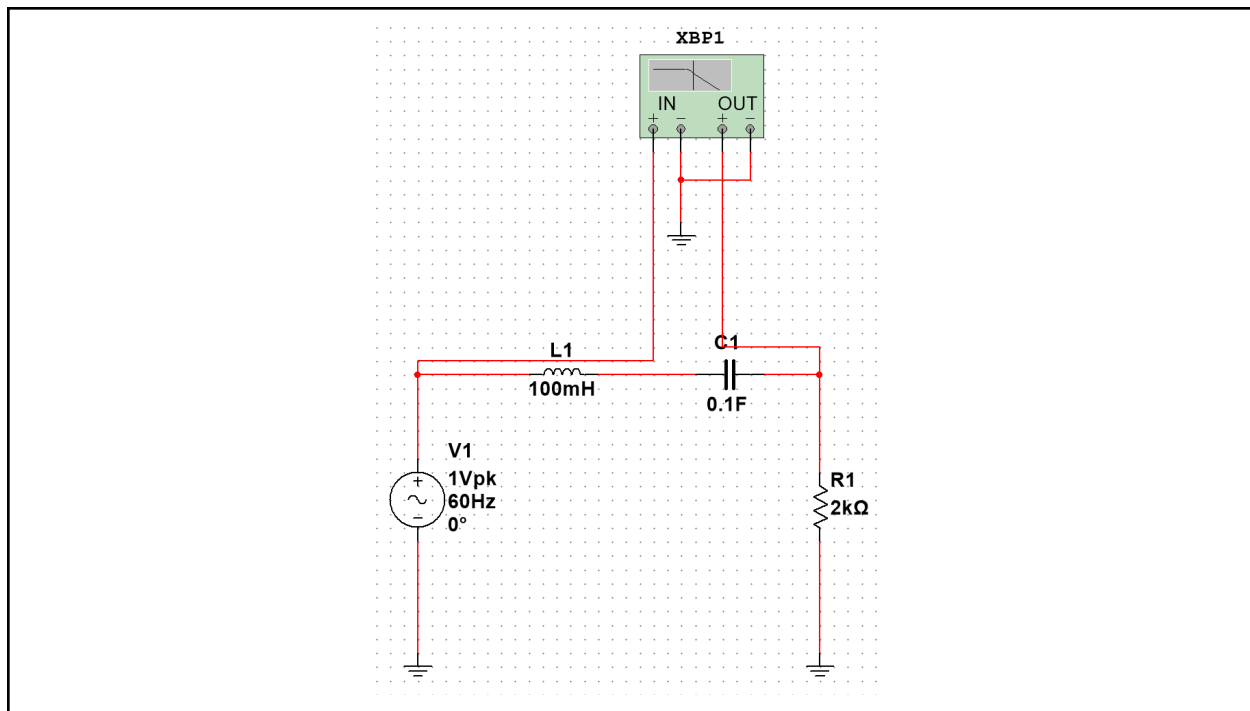
and HP to increase the efficiency of calculations

	Circuit from Multisim
L	100mH
C	0.1F
R (underdamped)	1,400 Ω
R (overdamped)	2,000 Ω
R (critically damped)	4,400 Ω

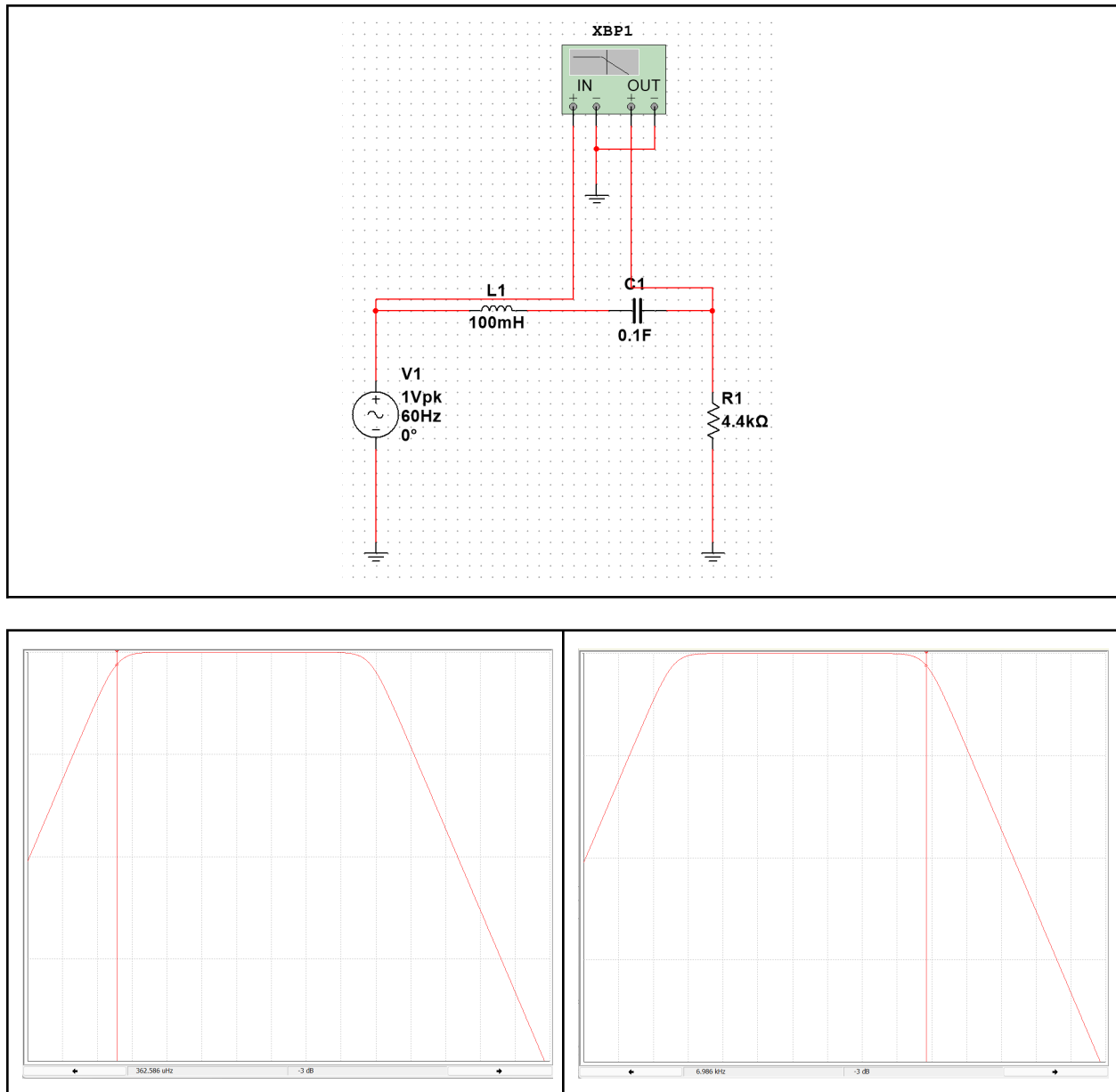
Table 20



IMG 15. Bode plot of the simulated circuit, for underdamped



IMG 16. Bode plot of the simulated circuit, for overdamped



IMG 17. Bode plot of the simulated circuit, for critically damped

Calculations and analysis:

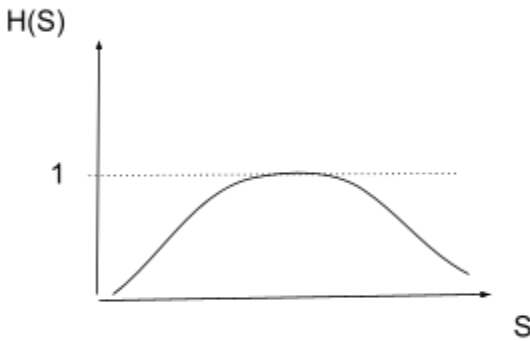
1. Filter behavior

Note that we have three different resistance values, but the behavior of the circuit stays the same and we can simply leave the variable $\frac{R}{L}$ in place and not plug in a specific resistance value.

$$H(S) = \frac{\frac{R}{L}S}{S^2 + (\frac{R}{L})S + (\frac{1}{LC})} = \frac{\frac{R}{L}S}{S^2 + \frac{R}{L}S + \frac{1}{(100mH)(0.1uF)}} = \frac{\frac{R}{L}S}{S^2 + (\frac{R}{L})S + (10^8)}$$

$$\text{As } S \rightarrow 0, H(s) = \frac{\frac{R}{L}S}{S^2 + (\frac{R}{L})S + (10^8)} \approx \frac{\frac{R}{L}(0)}{(0)^2 + (\frac{R}{L})(0) + (10^8)} \approx \frac{0}{10^8} \approx 0$$

$$\text{As } S \rightarrow \infty, H(s) = \frac{\frac{R}{L}S}{S^2 + (\frac{R}{L})S + (10^8)} \approx \frac{\frac{R}{L}(\infty)}{(\infty)^2 + (\frac{R}{L})(\infty) + (10^8)} \approx \frac{\infty}{\infty^2 + \infty + 10^8} \approx 0$$



Based on this analysis of the limits of s , the circuit attenuates low and high frequency signals thus **behaving like a band pass filter**

2. Stability and filter behavior analysis:

2a. Underdampened:

$$R=1400\Omega,$$

$$H(s) = \frac{\frac{R}{L}S}{S^2 + (\frac{R}{L})S + (10^8)} = \frac{\frac{1400}{(100mH)}S}{S^2 + \frac{1400}{100mH}S + \frac{1}{(100mH)(0.1uF)}} = \frac{14000S}{S^2 + (14,000)S + (10^8)} = \frac{n(Z)}{n(P)}$$

$$n(Z) = 14,000s, \text{ if } n(Z) = 0, s = 0$$

Since $s=0$, we have a single zero at zero, which indicates band pass behavior (as referenced in table 1)

$$n(p) = s^2 + 14,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 14,000$$

$$c = 10^8$$

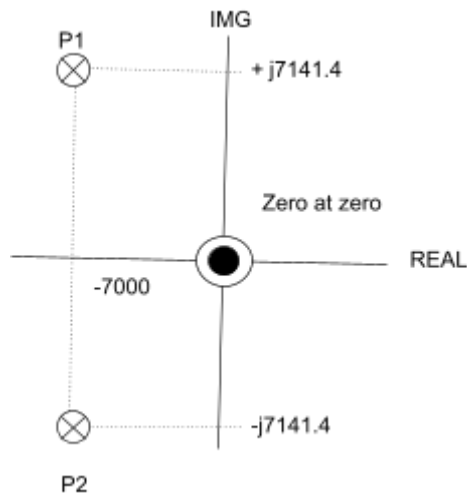
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(14,000) \pm \sqrt{(14,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -7000 \pm \frac{\sqrt{-204 \cdot 10^8}}{2} \rightarrow$$

$$-7000 \pm j7141.4$$

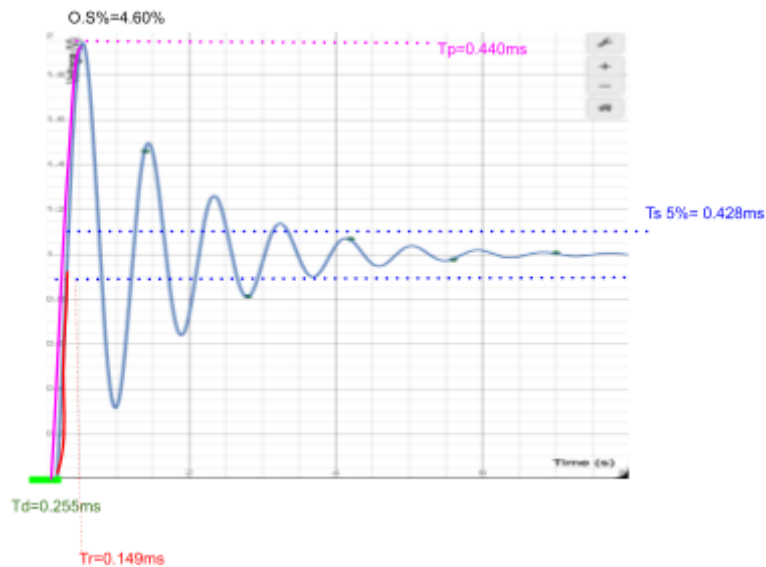
Two negative, complex poles. We have two negative poles, and the system is stable.

$$P_1 = -7000 + j7141.4$$

$$P_2 = -7000 - j7141.4$$



IMG 18. Oscillating behavior is seen



IMG 19. Please note that this image is a rough estimation and is based on image 9.

	Formula	Calculation	Result
Dampening factor	$\beta = \sqrt{1 - \xi^2}$	$\beta = \sqrt{1 - 0.7^2}$	$\beta = 0.7141$

	$\omega_n = \frac{1}{\sqrt{LC}}$	$\omega_n = \frac{1}{\sqrt{(0.1H)(1*10^{-7}F)}}$	$\omega_n = 10,000 \text{ rad/s}$
	$\omega_d = \beta\omega_n$	$\omega_d = (0.7141)(10,000 \text{ rad/s})$	$\omega_d = 7141 \text{ rad/s}$
Rise time	$T_R = \frac{1+0.7\xi}{\omega_n}$	$T_R = \frac{1+0.7(0.7)}{10,000}$	$T_R = 0.149\text{ms}$
Delay time	$T_D = \frac{0.8+2.5\xi}{\omega_n}$	$T_D = \frac{0.8+2.5(0.7)}{10,000}$	$T_D = 0.255\text{ms}$
Peak time	$T_p = \frac{\pi}{\omega_d}$	$T_p = \frac{\pi}{7141}$	$T_p = 0.440\text{ms}$
% Overshoot	$O.S\% = e^{\frac{-\pi\xi}{\beta}} * 100$	$O.S\% = e^{\frac{-\pi 0.7}{0.7141}} * 100$	$O.S\% = 4.60\%$
Settling time	$t_s = \frac{-\ln(\% \text{ in decimal})}{\xi\omega_n}$	For 2%: $t_s = \frac{-\ln(0.02)}{(0.7)(10,000)}$	$t_s = 0.559\text{ms}$
		For 4%: $t_s = \frac{-\ln(0.04)}{(0.7)(10,000)}$	$t_s = 0.460\text{ms}$
		For 5%: $t_s = \frac{-\ln(0.05)}{(0.7)(10,000)}$	$t_s = 0.428\text{ms}$

Table 15.

The step response for an underdamped system is $V_o(t) = 1 - e^{-\beta t}(\cos(\omega_d t) + \frac{\xi}{\beta} \sin(\omega_d t))$

$$V_o(t) = 1 - e^{-(0.7141)t}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

From the standard underdamped form, $e^{-\beta t}$ so $-\beta t = -\frac{t}{\tau}$ and $\tau = \frac{1}{\beta} = \frac{1}{0.7141} = 1.40$

For 5τ , $5\tau = 5(1.4004) = 7.002$

$$\text{At } t = 5\tau, e^{-\beta t} = e^{-\beta(5\tau)} = e^{-\beta(5\frac{1}{\beta})} = e^{-5}$$

$$V_o(t) = 1 - e^{-5}(\cos 7141t) + \frac{0.7}{0.7141} \sin(7141t)$$

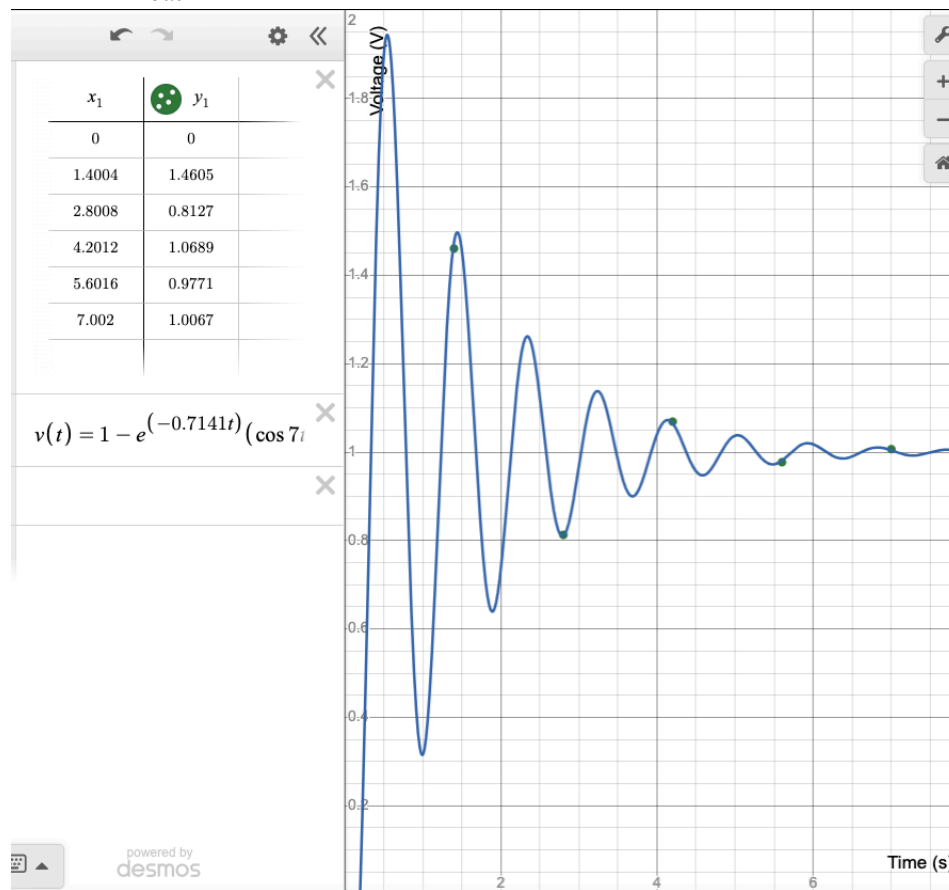
$$V_o(t) = 1 - 0.0067370(\cos(7141 * 7.002) + 0.9803 \sin(7141 * 7.002)) = 1.0067v$$

To find the % of the final value that the output is, $\frac{1.0067v}{1v} * 100 = 100.67\%$

	$V_{out}(t)$	% Final
0τ	0v	0%

$1\tau = 1.4004s$	1.461v	146.05%
$2\tau = 2.8008s$	0.8127v	81.27%
$3\tau = 4.2012s$	1.0689v	106.89%
$4\tau = 5.6016s$	0.9771v	97.71%
$5\tau = 7.0020s$	1.006v	100.67%

Table 16. $V_{out}(t)$ after time constants



IMG 20. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was divided by 1000 to decrease the frequency to make the points visible with the line of best fit. The equation used in this graph is $V_o(t) = 1 - e^{-(0.7141)t} (\cos(7t) + \frac{0.7}{0.7141} \sin(7t))$ instead of $V_o(t) = 1 - e^{-(0.7141)t} (\cos(7141t) + \frac{0.7}{0.7141} \sin(7141t))$

2b. Critically damped:

$$R = 2000\Omega, \text{ so } H(s) = \frac{\frac{R}{L}S}{s^2 + (\frac{R}{L})S + (10^8)} = \frac{\frac{2000}{(100mH)}S}{s^2 + \frac{2000}{100mH}S + \frac{1}{(100mH)(0.1uF)}} = \frac{20,000s}{s^2 + (20,000)s + (10^8)} = \frac{n(Z)}{n(P)}$$

$$n(Z) = 20,000, \text{ if } n(Z) = 0, s = 0$$

Since $s=0$, we have a single zero at zero, which indicates band pass behavior (as referenced in table 1)

$$n(p) = s^2 + 20,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

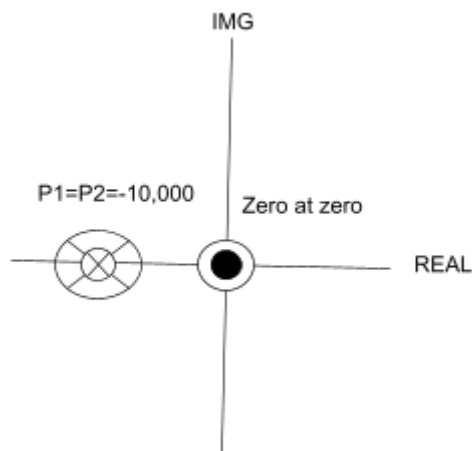
$$b = 20,000$$

$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(20,000) \pm \sqrt{(20,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -10,000 \pm \frac{\sqrt{40,000 - 40,000}}{2} \rightarrow -10,000 \pm \frac{\sqrt{0}}{2}$$

$$P_1 = P_2 = -10,000$$

Two identical, negative, real poles. We have two negative poles, and the system is stable.



IMG 21.

The step response for an underdamped system is $V_o(t) = 1 - [e^{-pt} + te^{-pt}]u(t)$

$$V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$$

$$e^{-pt} \text{ so } -pt = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{p} = \frac{1}{10,000} = 0.0001s = 0.1ms$$

$$\text{For } 5\tau, 5\tau = 5(0.1ms) = 0.0005ms$$

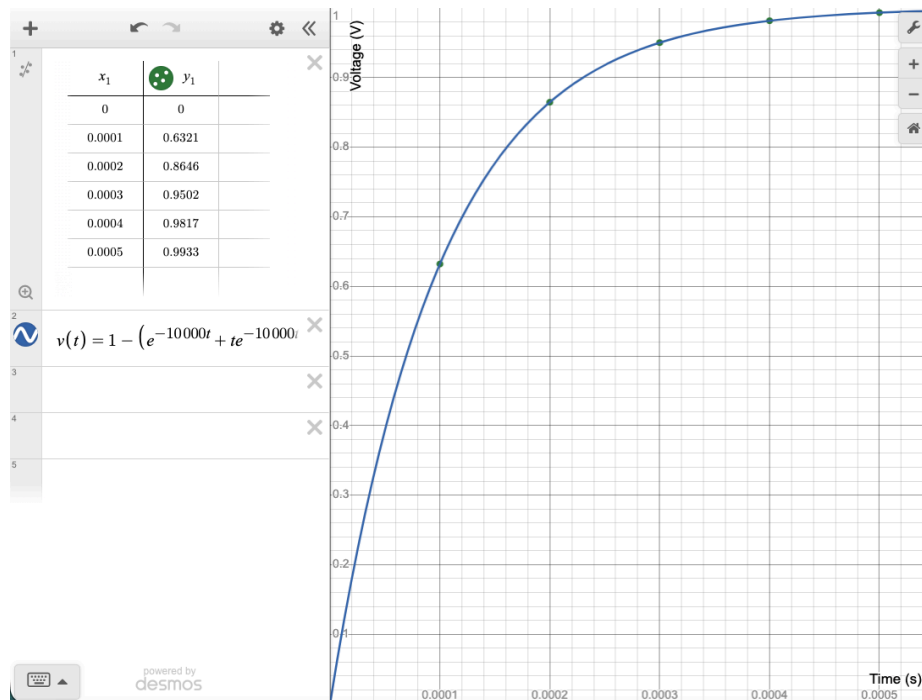
$$\text{At } t = 5\tau, e^{-pt} = e^{-p(5\tau)} = e^{-p(5\frac{1}{p})} = e^{-5}$$

$$V_o(t) = 1 - [e^{-5} + te^{-5}]u(t) = 1 - [e^{-5} + 0.0005e^{-5}]u(t) = 0.9933$$

$$\text{To find the \% of the final value that the output is, } \frac{0.9933v}{1v} * 100 = 99.33\%$$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 0.1ms$	0.6321v	63.21%
$2\tau = 0.2ms$	0.8646v	81.27%
$3\tau = 0.3ms$	0.9502v	95.02%
$4\tau = 0.4ms$	0.9817v	98.17%
$5\tau = 0.5ms$	0.9933v	99.33%

Table 17. $V_{out}(t)$ after time constants



IMG 22. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com/). Note that the equation of the graph was $V_o(t) = 1 - [e^{-10,000t} + te^{-10,000t}]u(t)$

2c. Overdamped

$$R = 4400\Omega, \text{ so } H(s) = \frac{\frac{R}{L}s}{s^2 + (\frac{R}{L})s + (10^8)} = \frac{\frac{4400}{(100mH)}s}{s^2 + \frac{2000}{100mH}s + \frac{1}{(100mH)(0.1uF)}} = \frac{44,000s}{s^2 + (44,000)s + (10^8)} = \frac{n(Z)}{n(P)}$$

$$n(Z) = 44,000s, \text{ if } n(Z) = 0, s = 0$$

Since $s=0$, we have a single zero at zero, which indicates band pass behavior (as referenced in table 1)

$$n(p) = s^2 + 40,000s + 10^8$$

To set $n(p) = 0$, we must use the quadratic equation to find the roots

$$a = 1$$

$$b = 40,000$$

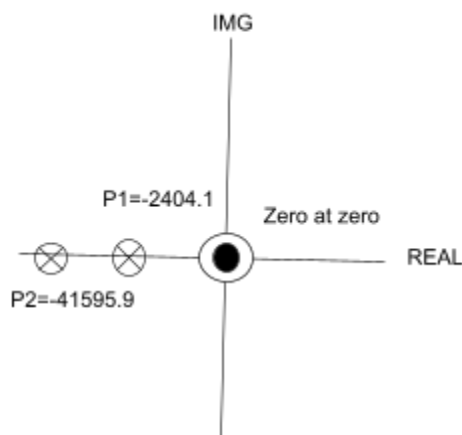
$$c = 10^8$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(40,000) \pm \sqrt{(40,000)^2 - 4(1)(10^8)}}{2(1)} \rightarrow -20,000 \pm \frac{39191.8}{2}$$

$$P_1 = -2404.1$$

$$P_2 = -41595.9$$

Two different, negative, real poles. We have two negative poles, and the system is stable.



IMG 23.

The step response for an overdamped system is $V_o(t) = [1 - (Ae^{-p_1t} + Be^{-p_2t})]u(t)$

$$V_o(t) = [1 - (Ae^{-2404.1t} + Be^{-41595.9t})]u(t)$$

Using partial fractions: $\frac{n(s)}{(s+A)(s+B)} = \frac{A}{s+a} + \frac{B}{s+b}$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A}{(s+2404.1)} + \frac{B}{(s+41595.9)}$$

$$\frac{s}{(s+2404.1)(s+41595.9)} = \frac{A(s+41595.9)}{(s+2404.1)(s+41595.9)} + \frac{B(s+2404.1)}{(s+2404.1)(s+41595.9)}$$

$$n(s) = A(s + 41595.9) + B(s + 2404.1)$$

Expand: $n(s) = As + A41595.9 + Bs + B2404.1$

Regroup: $n(s) = (A + B)s + A41595.9 + B2404.1$

$$(A + B)s = (1)s, A + B = 1, B = 1 - A$$

$$A41595.9 + B2404.1 = 0, A41595.9 + (1 - A)2404.1 = 0, (41595.9 - 2404.1)A + 2404.1 = 0$$

$$39191.8A + 2404.1 = 0, A = -0.0614 \text{ and } B = 1.0614.$$

Therefore, the step response is $V_o(t) = [1 - (1.0614e^{-2404.1t} + -0.0614e^{-41595.9t})]u(t)$

We need to decide which exponent to use. There are two exponentials, so technically two time constants. I choose to use the smaller pole magnitude because it decays slowly and controls the overall behavior more.

$$e^{-2404.1t} \text{ so } -2404t = -\frac{t}{\tau} \text{ and } \tau = \frac{1}{2404.1} = 0.4160 \text{ ms}$$

$$\text{For } 5\tau, 5\tau = 5(0.4160 \text{ ms}) = 2.07980 \text{ ms}$$

$$\text{At } t = 5\tau, e^{-2404.1t} = e^{-2404.1(5\tau)} = e^{-2404.1(5\frac{1}{2404.1})} = e^{-5}$$

$$V_o(t) = [1 - (1.0614e^{-5} + -0.0614e^{-41595.9(2.07980 \text{ ms})})]u(t) = 0.992848$$

$$\text{To find the \% of the final value that the output is, } \frac{0.9928v}{1v} * 100 = 99.28\%$$

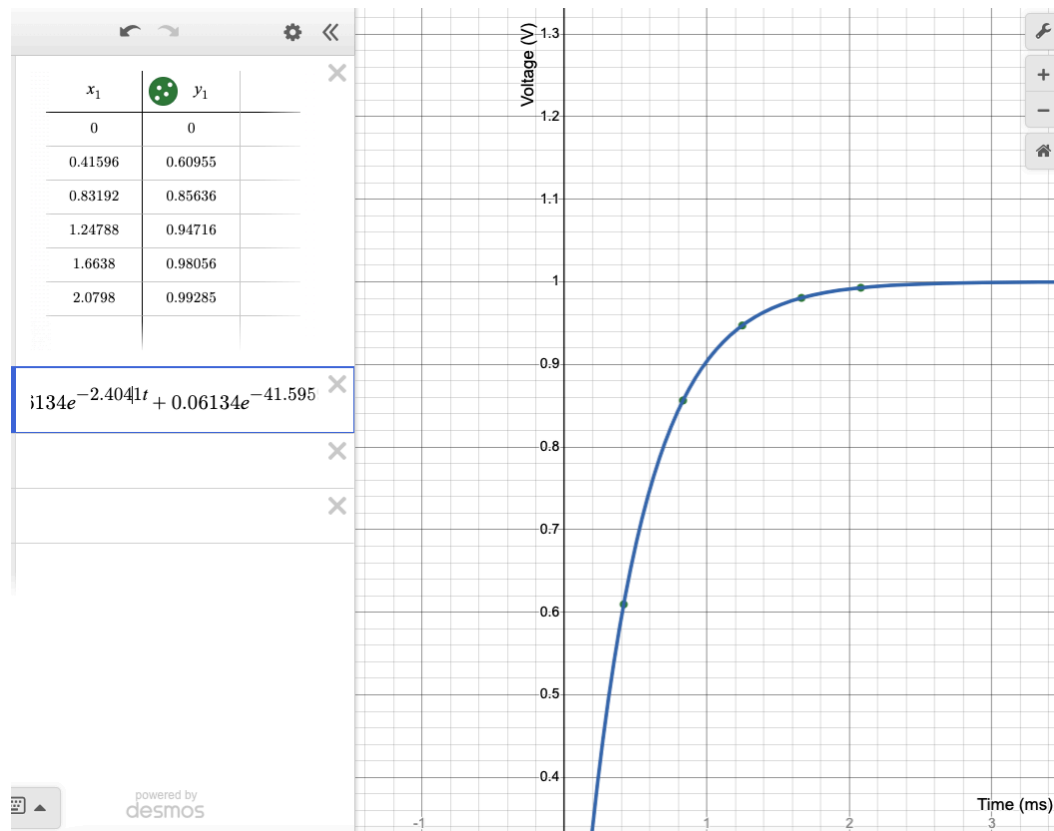
Recall that for critically damped, at 5τ (0.5 ms) there was $0.9933v$ or 99.33% of the final

voltage. It tracks that for overdamped at 5τ (2.07 ms) there is slightly less voltage ($0.9928v$)

and 99.28% of the final voltage, showing a slower rise time and less voltage.

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 0.416ms$	0.6096v	60.96%
$2\tau = 0.8319ms$	0.8564v	85.64%
$3\tau = 1.2479ms$	0.9471v	94.71%
$4\tau = 1.6638ms$	0.9805v	98.05%
$5\tau = 2.0798ms$	0.9928v	99.28%

Table 18. $V_{out}(t)$ after time constants



IMG 24. Steady state is reached after 5τ , graphed using [Desmos.com](https://desmos.com). Note that the equation of

the graph was $V_o(t) = 1 - (1.0614e^{-2.4041t} + -0.06134e^{-41.5959t})$

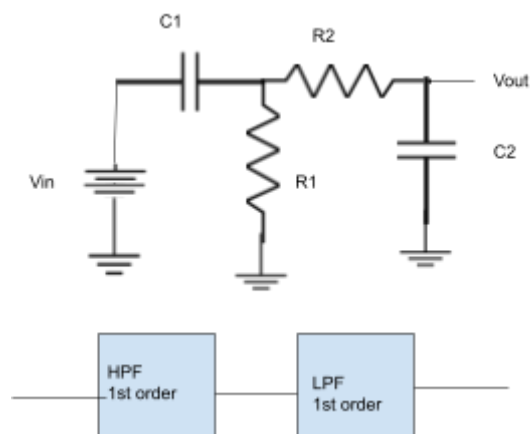
Results:

It was determined that the resistance for underdamped was 1400Ω , for critically damped 2000Ω and for over damped 4400Ω . These resistance values were reused from the high and low pass filter calculation and analysis portion, for simplification. As a result, the process and many results were quite similar. This was done intentionally, for efficiency purposes.

Analysis of the transfer function using the limits of s showed that the circuit attenuates high frequencies and also attenuates low frequencies, confirming that it behaves as a band pass filter. For the case of underdamped, it is found that there are two negative complex poles (indicating stability), and we have one zero at zero (confirming band pass filter behavior and matching the content in table 1). The oscillatory response observed in both the real-time graph and the $v_{out}(t)$ versus τ is consistent with expected oscillatory behavior. Additionally, the calculated percent overshoot of 4.60% tracks well with the oscillating behavior, and all three of the settling time calculations correspond when observing the graph in image 9. The exact points of T_R , T_D , T_p , t_s can be found in the graph in image 9 to confirm the calculations and accuracy of the formulas. Overall, the underdamped response matches the expected oscillatory behavior of a second-order passive filter.

For critically damped, we see that we have two identical negative real poles, indicating a stable system, and a zero at zero (confirming band pass behavior). For overdamped we have two different, negative, real poles with a zero at zero (again confirming bandpass behavior). For critically damped, at 5τ ($0.5ms$) there was $0.9933v$ or 99.33% of the final voltage. It tracks that for overdamped at 5τ ($2.07ms$) there is slightly less voltage ($0.9928v$) and 99.28% of the final voltage, showing a *slower* rise time and less voltage.

2 loop passive second order RC filter:



When choosing values for R and C, we need to make sure that the first filter is a high pass and the second is a low pass ($F_{c1} < F_{c2}$)

Let $C1=C2=0.1\mu F$, but $R_1 \neq R_2$ in order to make two different frequencies.

Increasing R decreases the frequency, so R2 must be larger than R1

$$XC = R, \quad \frac{1}{\omega_c C} = R, \quad \frac{1}{2\pi f_c C} = R, \quad f_c = \frac{1}{R\pi 2C}$$

$$f_{c2} = \frac{1}{R\pi 2C} = \frac{1}{(220\Omega)(2\pi)(1*10^{-6}f)} = 723.4\text{hz}$$

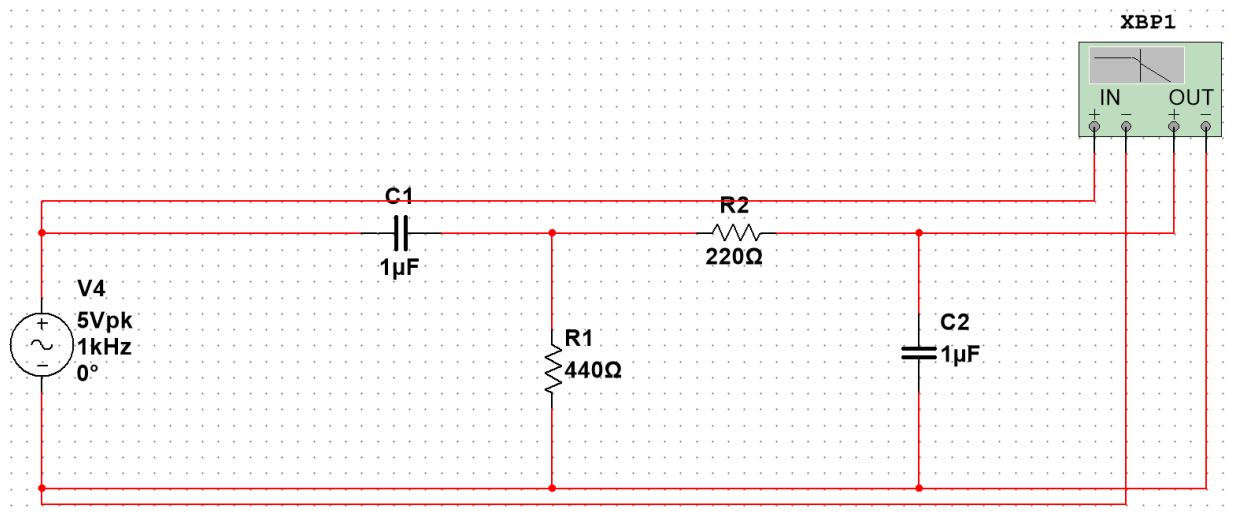
$$f_{c1} = \frac{1}{R\pi 2C} = \frac{1}{(440\Omega)(2\pi)(1*10^{-6}f)} = 361.7\text{hz}$$

$$\omega_o = \sqrt{\omega_1 \omega_2} \text{ so } f_o = \sqrt{f_1 f_2} = \sqrt{361.7 * 723.4} = 511.5$$

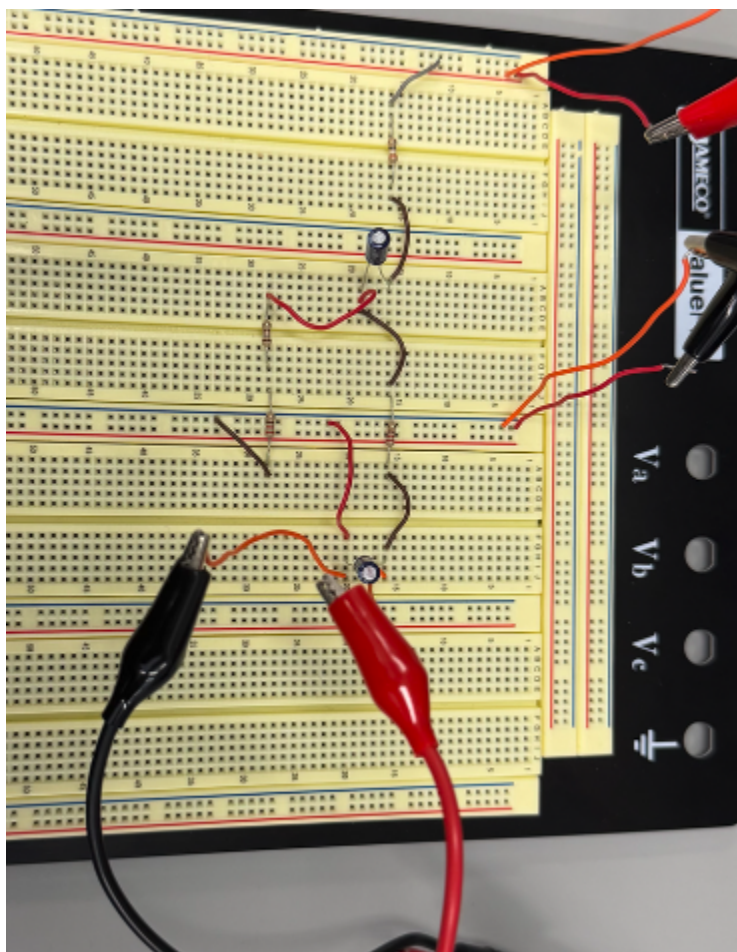
Bandwidth is approximately $f_{c2} - f_{c1} = 723.4 - 361.7 = 361.7\text{hz}$

	Multisim	breadboard
R1	440Ω	439Ω
C1	1uF	1uF
R2	220Ω	220Ω
C2	1uF	1uF

Table 19.

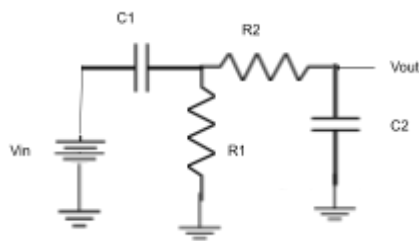


IMG 25. Two loop passive second order filter. Note that the bode plot is not shown due to errors caused by Multisim's measurement system.



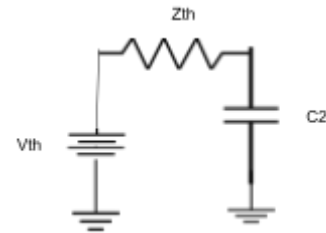
IMG 26. Physical circuit on the breadboard of the two-loop filter circuit.

Calculations:



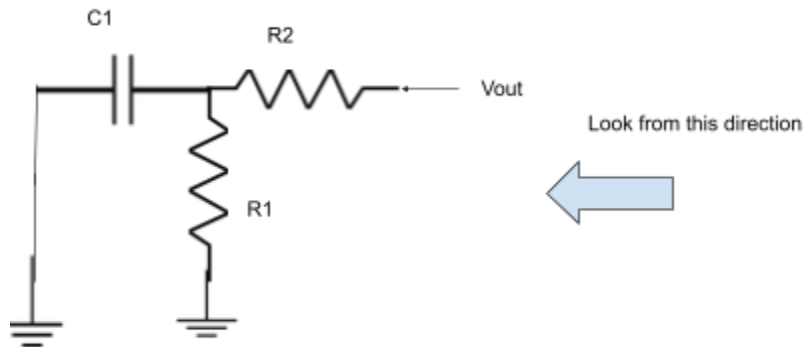
Current state:

goal end state:



Recall $C1=C2=C$

1) Find Z_{th}

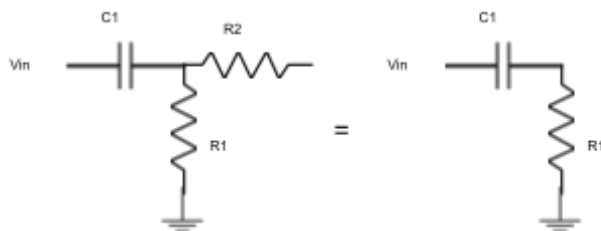


The voltage source is removed as well as $C2$.

$$Z_{th} = R2 + R1 // \frac{1}{Cs} = R2 + \frac{R1 * \frac{1}{Cs}}{R1 + \frac{1}{Cs}} = \frac{R1}{R1Cs + 1} + R2 = \frac{R1 + R2(R1Cs + 1)}{R1Cs + 1} = \frac{R1 + R2R1Cs + R2}{R1Cs + 1} = \frac{R1 + R2 + R1R2Cs}{R1Cs + 1}$$

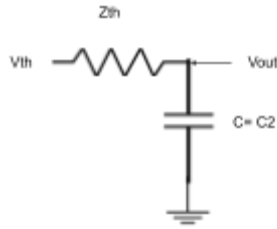
2) Find V_{th}

$R2$ can be disregarded because it is connected to an open so essentially has infinite resistance, and no current will take that path.



$$V_{th} = \frac{R1}{R1 + \frac{1}{Cs}} Vin = \frac{R1Cs}{R1Cs + 1} Vin$$

3) Find V_{out} and thus $H(S)$



$$V_{out} = V \text{ across capacitor} = \frac{\frac{1}{CS}}{\frac{1}{CS} + Z_{th}} v_{th} = \frac{\frac{1}{CS}}{\frac{1}{CS} + \frac{R1}{R1CS+1}} v_{th}$$

$$V_{th} = \frac{R1CS}{R1CS+1} V_{in}$$

$$H(S) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + \frac{R1+R2+R1R2CS}{R1CS+1}} \frac{V_{th}}{V_{in}} =$$

$$\frac{\frac{1}{CS}}{\frac{1}{CS} + \frac{R1+R2+R1R2CS}{R1CS+1}} * \frac{R1CS}{R1CS+1} * \frac{CS}{CS} = \frac{\frac{1}{R2C} S}{R1R2C^2 S^2 + \frac{(R1+R2)CS+R1CS+1}{(2R1+R2)CS}}$$

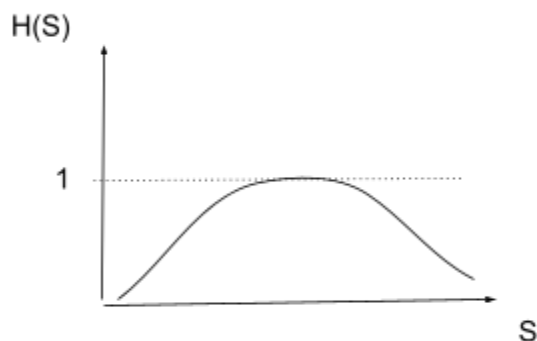
$$H(S) = \frac{(\frac{1}{R2C})S}{S^2 + (\frac{2R1+R2}{R1R2C})S + \frac{1}{R1R2C^2}}$$

2. Now we can observe filter behavior

$$\text{As } S \rightarrow 0, H(S) = \frac{(\frac{1}{R2C})S}{S^2 + (\frac{2R1+R2}{R1R2C})S + \frac{1}{R1R2C^2}} \approx \frac{0}{(0)^2 + (0) + \frac{1}{440*220*1*10^{-6}}} \approx \frac{0}{10.33} \approx 0$$

As $S \rightarrow \infty$,

$$H(s) = \frac{(\frac{1}{R2C})S}{S^2 + (\frac{2R1+R2}{R1R2C})S + \frac{1}{R1R2C^2}} \approx \frac{\infty}{(\infty)^2 + (\infty) + \frac{1}{440*220*1*10^{-6}}} \approx 0 \text{ after using L'Hopital's rule}$$



Based on this analysis of the limits of s , the circuit attenuates low and high frequency signals

thus **behaving like a band pass filter**

2. Stability and filter behavior analysis:

We can compare to the standard second order equation to find the values of components.

$$\frac{(\frac{1}{R2C})S}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{ to } \frac{(\frac{1}{R2C})S}{s^2 + (\frac{2R1+R2}{R1R2C})S + \frac{1}{R1R2C2}}$$

$$\text{Therefore } \frac{2R1+R2}{R1R2C} = 2\xi\omega_n \text{ and } \omega_n^2 = \frac{1}{R1R2C2}, \text{ so } \omega_n = \frac{1}{\sqrt{R1R2C}} =$$

We can determine if the circuit is under, over or critically dampened by solving for the dampening ratio.

$$\frac{2R1+R2}{R1R2C} = 2\xi\omega_n, \text{ so } \frac{2R1+R2}{R1R2C} 2\omega_n = \xi, \text{ so } \frac{2(440)+(220)}{440*220*1*10^{-6}} 2(3214.12 \text{ rad/s}) = \xi$$

$$\xi = 1.768$$

Therefore this is an overdamped circuit.

$$H(S) = \frac{(\frac{1}{R2C})S}{s^2 + (\frac{2R1+R2}{R1R2C})S + \frac{1}{R1R2C2}} = \frac{4545.5s}{s^2 + 11363.64s + 1.03306*10^7} = \frac{n(Z)}{n(P)}$$

$$n(Z) = 4545s, \text{ if } n(Z) = 0, s = 0$$

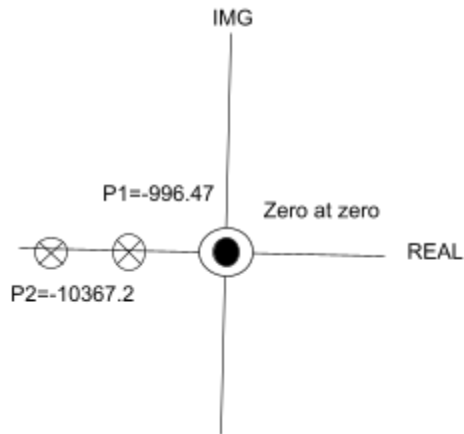
Since $s=0$, we have a single zero at zero, which indicates band pass behavior (as referenced in table 1)

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(11363.64) \pm \sqrt{(11363.64)^2 - 4(1)(1.03*10^7)}}{2(1)}$$

$$P_1 = -996.47$$

$$P_2 = -10367.17$$

Two different, negative, real poles. We have two negative poles, and the system is stable.



The step response for an overdamped system is $V_o(t) = [1 - (Ae^{-p_1t} + Be^{-p_2t})]u(t)$

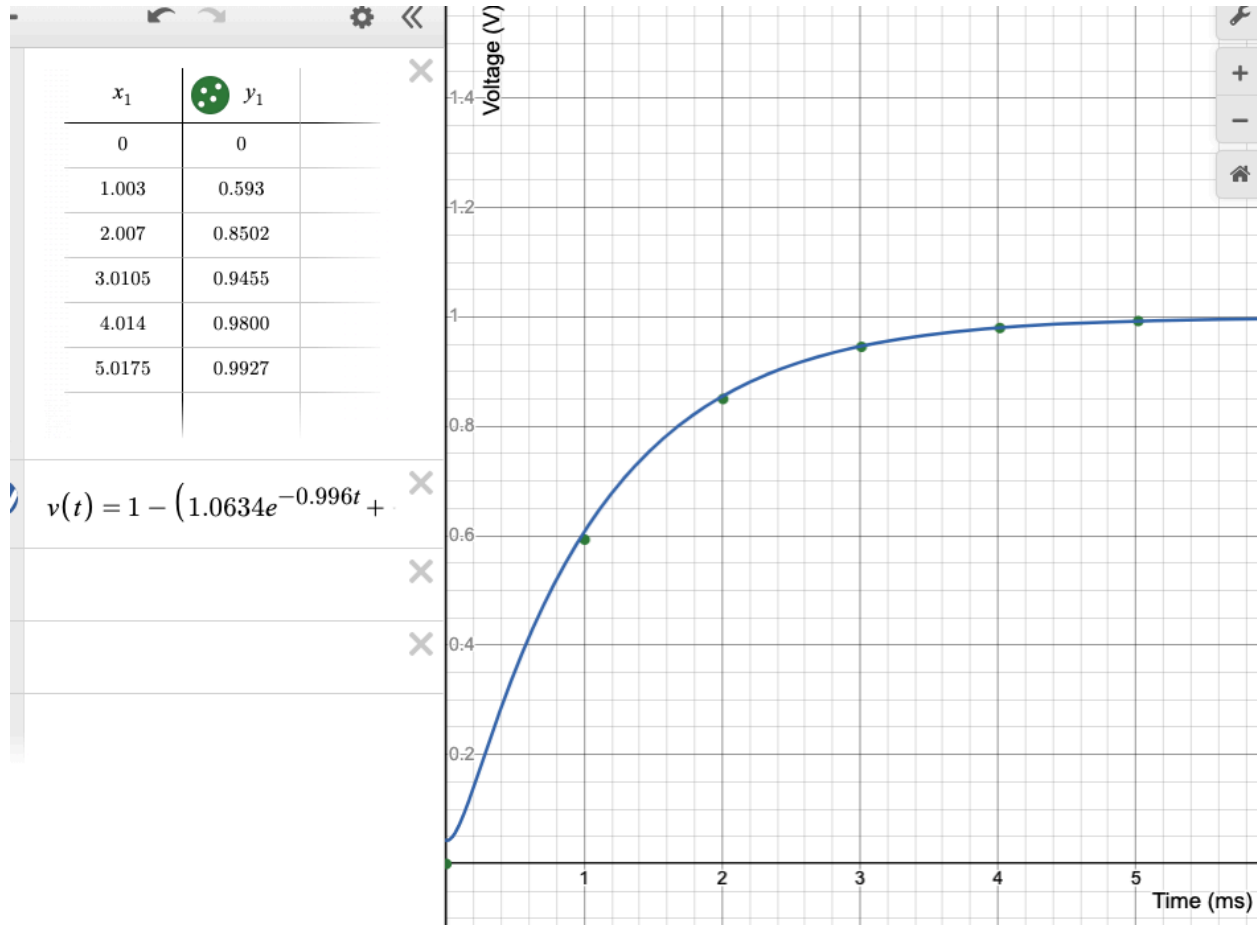
$$V_o(t) = [1 - (Ae^{-996.47t} + Be^{-10367.2t})]u(t)$$

$$\text{Or } V_o(t) = [1 - (1.1063e^{-996.47t} + -0.1063e^{-10367.17t})]u(t)$$

$$\tau = \frac{1}{996.47} = 1.0035ms$$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 1.0035ms$	0.5930v	59.3%
$2\tau = 2.0070ms$	0.8502v	85.02%
$3\tau = 3.0105ms$	0.9455	94.55%
$4\tau = 4.0140ms$	0.9800	98.00%
$5\tau = 5.0175ms$	0.9927v	99.27%

Table 20. $V_{out}(t)$ after time constants



IMG 27. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com). Note that the equation of the graph was $V_o(t) = 1 - (1.0634e^{-0.99647t} + -0.10634e^{-10.36717t})$

Discussion

This experiment analyzed second-order passive filters by comparing theoretical predictions, Multisim simulations, and experimental results. Overall, the circuits demonstrated the expected frequency response behavior, confirming the validity of the derived transfer functions. While the exact numerical agreement was not perfect, the general trends and shapes of the responses were consistent across all methods.

For the low-pass and high-pass filters, the observed behavior aligned with theory under different damping conditions. The underdamped case ($\zeta < 1$) showed slight resonance near the cutoff frequency, the critically damped case ($\zeta = 1$) produced a smooth transition, and the overdamped case ($\zeta > 1$) resulted in a more gradual response. The band-pass filter exhibited the expected rise to a peak at the center frequency followed by a decrease, while the band-stop filter showed a clear attenuation region. These trends confirm that the qualitative behavior of all filters matched theoretical expectations.

Although the Bode plots did not perfectly match in magnitude, the overall shapes and slopes were consistent, including the approximate -40 dB/decade roll-off for second-order systems. The characteristic curves such as the peak in the band-pass filter and attenuation regions were clearly observed, indicating correct circuit operation.

Possible sources of error include both experimental and simulation-related factors. Experimentally, discrepancies may arise from component tolerances, parasitic capacitance and inductance in the breadboard, loose connections, and measurement limitations of instruments. In addition, software-related factors in Multisim may also contribute to differences. These can include numerical approximations used by the simulator, default component models that may not perfectly match real components, simulation resolution settings (such as frequency sweep step size), and rounding errors in calculations. Improper configuration of simulation parameters or probe placement can also slightly affect results.

Despite these sources of error, the experiment successfully demonstrated the expected behavior of second-order filters. The agreement in overall response shape and slope confirms that both the

theoretical analysis and simulation tools are reliable, while also highlighting the importance of considering real-world and computational limitations when interpreting results.

Conclusion

This experiment successfully analyzed and verified the behavior of second-order passive filters, including low-pass, high-pass, band-pass, and band-stop configurations, through theoretical derivation, simulation, and experimental measurement. The results confirmed that filter performance is primarily determined by pole locations and the damping ratio (ζ), which define whether the system is underdamped, critically damped, or overdamped.

In all filter types, the underdamped case ($\zeta < 1$) produced resonance or peaking near the cutoff or center frequency, improving selectivity but introducing potential instability and overshoot. The critically damped condition ($\zeta = 1$) provided the most balanced response, with no overshoot and a smooth transition between passband and stopband, making it the most practical for stable applications. The overdamped case ($\zeta > 1$) resulted in a slower, more gradual response with reduced selectivity, confirming the trade-off between stability and sharpness.

The band-pass filter demonstrated how damping affects bandwidth and quality factor, where lower damping produced a narrower and selective passband. The band-stop filter further showed how specific frequency ranges can be effectively attenuated using multi-loop designs.

Across all configurations, the observed roll-off rate approached the expected -40 dB/decade, consistent with second-order systems. Measured cutoff frequencies and -3 dB points closely matched theoretical and simulated values, with minor differences due to component tolerances, parasitic effects, and experimental limitations.

Overall, this lab validated the fundamental principles of second-order filter design and highlighted key engineering trade-offs between selectivity, stability, and response speed.

References:

- **Circuit Analysis in Frequency Domain & Analog Filter Design**

M. Mousavi, *Circuit Analysis in Frequency Domain & Analog Filter Design*, The Electromechanical System Engineering Technology Department, California State Polytechnic University, Pomona.